Fisher-Glass Learning: Classification Bounds and Subpopulation Audits

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Abstract

High predictive accuracy can coexist with sensitivity to task-preserving nuisance changes. Building on the Fisher-glass framework, we derive local classification lower bounds and subpopulation audits based on nuisance-projected information. The bounds connect efficient task information to minimax excess classification risk under explicit distinguishability and decision-separation conditions. We derive regularization-aware audit bounds and evaluate subpopulation selection rules fixed before evaluation. On Waterbirds, an uncentered feature–Schur diagnostic adjusts task-logit moments for scores from a classifier trained to predict background type, selecting a group meeting validation-performance criteria. Across five training seeds, interventions preserving the extracted bird foreground and labels yield 93.2% accuracy when land/water background type is preserved and 40.6% when it is changed, a 52.6-percentage-point difference. Search using task labels and a fixed nuisance direction also identifies held-out populations with high accuracy and low estimated projected predictive Fisher information. Training and acquisition experiments evaluate changes in information and predictive performance together. The framework pairs predictive performance with explicit measures of task information and nuisance alignment.

1 Introduction

Reliable prediction under distribution shift requires understanding a model's evidence. A classifier can perform well using a nuisance cue correlated with its label, then fail when that correlation changes. Group-robust learning and error-based reweighting emphasize subpopulation performance [Sagawa et al., 2020, Liu et al., 2021]. We ask a complementary question: *how much task information remains after nuisance variation is accounted for?*

We investigate this question on Waterbirds, a bird-classification benchmark with correlated backgrounds. Among groups meeting validation-performance criteria, we select the smallest fraction of task-logit second moment remaining after ridge adjustment for a background-prediction score. We call this the *uncentered feature–Schur fraction*; the background probe is a classifier trained to predict land/water background type. Models and the selected group are frozen before held-out evaluation. Across five seeds, the group achieves 92.0% accuracy on untouched images. Background replacements preserving the extracted bird foreground and labels yield 93.2% accuracy when background type is preserved and 40.6% when it is changed: a controlled difference of 52.6 percentage points within an accurately classified population.

Efficient-score theory describes task information after nuisance adjustment through an orthogonal projection [Bickel et al., 1993, Tsiatis, 2006]. The Fisher-glass framework of Mansouri and Arai [2026] connects this geometry to inverse-information tails and a population tail certificate, describing how frequently nuisance-adjusted information becomes very small across environments. We study

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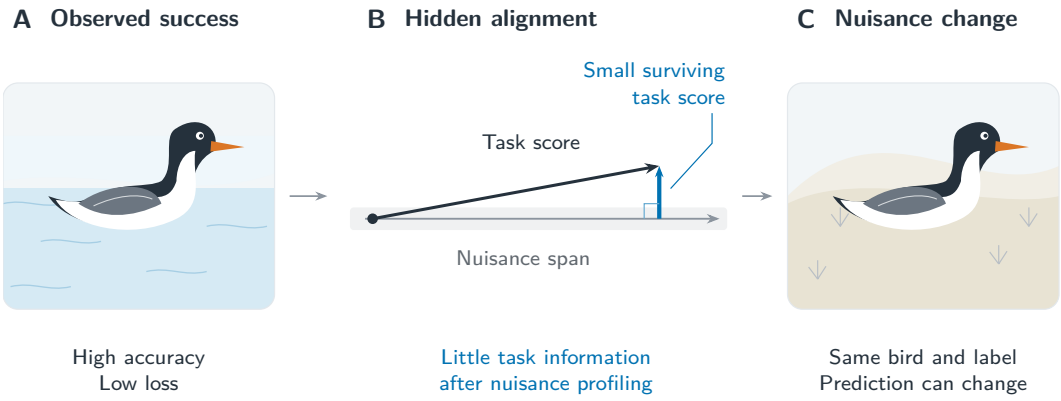


Figure 1: **Conceptual motivation for auditing task information under nuisance variation.** A nuisance cue can support high accuracy and low loss (A), while task–nuisance alignment leaves little information after projection (B). A task-preserving background change can expose the resulting sensitivity (C). The identical vector bird is reused in both environments; these are conceptual illustrations, not experimental images.

its implications for classification difficulty and trained-predictor evaluation. Figure 1 introduces the geometry; Section 2 specifies task and nuisance coordinates.

Our classification result combines local indistinguishability with quantitative Bayes-decision separation. The bound retains environment-dependent information, separation constants, and local radius. Audits use regularization-aware bounds, subpopulation selection rules with minimum-support constraints, and independent evaluation. Given task labels and a fixed nuisance direction, search identifies a held-out Waterbirds population with 98.8% membership-weighted accuracy and low estimated projected predictive Fisher information.

Our contributions are:

- **Local classification theory.** We derive an excess-classification-risk lower bound under explicit distinguishability and decision-separation conditions, connecting efficient task information to local learning difficulty.
- **Statistical audit methodology.** We derive bounds for regularized Schur estimates and independently evaluated population gates, accounting for membership normalization and support correction.
- **Machine-learning evaluation.** We study controlled background sensitivity, held-out discovery of accurate low-information populations, and the joint information–performance outcomes of targeted training and acquisition.

2 Task–Nuisance Geometry and Information Collapse

Let $e \sim \Pi$ index an observed group or latent subpopulation. The audit fixes the task contrast, admissible nuisance variations, and common task-coordinate normalization. Information describes the task distinction remaining after nuisance adjustment.

2.1 Task Information After Nuisance Profiling

Consider a local model $P_{\theta,\mu,e}$ with task parameter θ and nuisance parameter μ. A likelihood score measures the local change in log likelihood as a parameter varies. At $P_e = P_{0,0,e}$, assume differentiability in quadratic mean, with centered, square-integrable task-score vector $s_{T,e}$ and nuisance-score vector $s_{N,e}$. Let $\mathcal{T}_{N,e}$ be the closed span of admissible nuisance scores in $L^2(P_e)$, with inner product $\langle f, g\rangle_e = \mathbb{E}_{P_e}[fg]$, and let $\Pi_{\mathcal{T}_{N,e}}$ denote orthogonal projection onto that span.

The **efficient task score** and its information are

$$s^{\text{eff}}_{T,e} = s_{T,e} - \Pi_{\mathcal{T}_{N,e}} s_{T,e},$$
$$F_e := F^{\text{eff}}_{T|N,e} = \mathbb{E}_{P_e}[s^{\text{eff}}_{T,e}(s^{\text{eff}}_{T,e})^\top] = J_{TT,e} - J_{TN,e}J^{-1}_{NN,e}J_{NT,e}. \tag{1}$$

The projection acts componentwise. The Schur expression applies when finite nuisance scores span $\mathcal{T}_{N,e}$ and $J_{NN,e} \succ 0$, with $J_{AB,e} = \mathbb{E}_{P_e}[s_{A,e}s_{B,e}^\top]$ [Bickel et al., 1993, Tsiatis, 2006, Cox and Reid, 1987]. Write $\lambda_e = \lambda_{\min}(F_e)$ for the weakest task information. A zero eigenvalue identifies a task direction contained in the nuisance span.

For scalar task and nuisance scores with positive raw information, their Fisher correlation $\rho_e = J_{TN,e}/\sqrt{J_{TT,e}J_{NN,e}}$ gives

$$F_e = J_{TT,e}(1-\rho_e^2), \qquad \underbrace{F_e/J_{TT,e}}_{\text{surviving-information fraction}} = 1-\rho_e^2. \tag{2}$$

Absolute information measures remaining curvature in the declared coordinates. The fraction measures the proportion retained, lies in $[0,1]$, and is invariant to nonzero scalar rescaling of either score. It compares alignment across slices with different raw information; its feature–Schur analogue supplies the prospective selection statistic. The Fisher-Glass certificate summarizes absolute information across environments.

2.2 Task–Nuisance Alignment and Loss–Information Separation

Let b_e, r_e be centered, orthogonal, unit-norm scores, with nuisance span $\text{span}\{b_e\}$. For $\delta_e \geq 0$, define the task score $a_e = b_e + \delta_e r_e$. Then

$$a_e = b_e + \delta_e r_e, \quad J_{TT,e} = 1+\delta_e^2, \quad F_e = \delta_e^2, \quad \frac{F_e}{J_{TT,e}} = \frac{\delta_e^2}{1+\delta_e^2}. \tag{3}$$

Projection removes b_e, leaving $\delta_e r_e$. Raw information stays at least one as efficient information vanishes quadratically; δ_e measures distance from the nuisance span. When nuisance features support low observed risk, low loss coexists with arbitrarily weak efficient information, even in accurately classified populations. Appendix A.1 gives the construction and proof. Figure 2 illustrates the population consequence: frequent near-alignment of task scores with nuisance spans produces a heavy inverse-information tail.

3 Information Tails and Local Classification Difficulty

The frequency of near-null environments determines the inverse-information tail. Its classification consequence additionally depends on how weak task alternatives change the Bayes decision.

3.1 Collapse Geometry Determines Information Tails

In the tangency construction, $\lambda_e = \delta_e^2$: if $\Pr(\delta_e < t) \sim Ct^\gamma$, then $\Pr(\lambda_e^{-1} > x) \sim Cx^{-\gamma/2}$. The geometric tail law in Theorem 1 relates the inverse-information exponent to null-set codimension, information vanishing order, and nearby probability mass.

Theorem 1 (Geometric tail law [Mansouri and Arai, 2026])**.** *Let $z \in \mathbb{R}^d$ parameterize environments with density p, and let the Fisher-null set $\mathcal{Z} = \{z : \lambda(z) = 0\}$ be a nonempty compact C^2 embedded submanifold without boundary, of codimension $c \in \{1,\dots,d\}$. Suppose λ is continuous on the closure of a fixed tubular neighborhood of $\mathcal{Z}$ and bounded away from zero outside it. In normal coordinates $z = s + r\omega$, let $s \in \mathcal{Z}$, r be normal distance, and ω a unit normal direction. As $r \downarrow 0$, assume uniformly that information and density have the respective asymptotic forms $a(s,\omega)r^{2m}(1+o(1))$ and $b(s,\omega)r^\nu(1+o(1))$, where $m > 0$, $\nu > -c$, and the continuous coefficients are uniformly positive and bounded. Then, for some $C > 0$,*

$$\Pr(\lambda(z)^{-1} > x) \sim Cx^{-\beta}, \qquad \beta = \frac{c+\nu}{2m}. \tag{4}$$

An information threshold defines a narrow tube around $\mathcal{Z}$; integrating its probability mass determines the exponent. Greater codimension makes simultaneous collapse rarer, whereas higher-order vanishing produces a heavier tail. Quadratic collapse with positive density gives $\beta = c/2$, and $\mathbb{E}[\lambda_e^{-1}]$ is

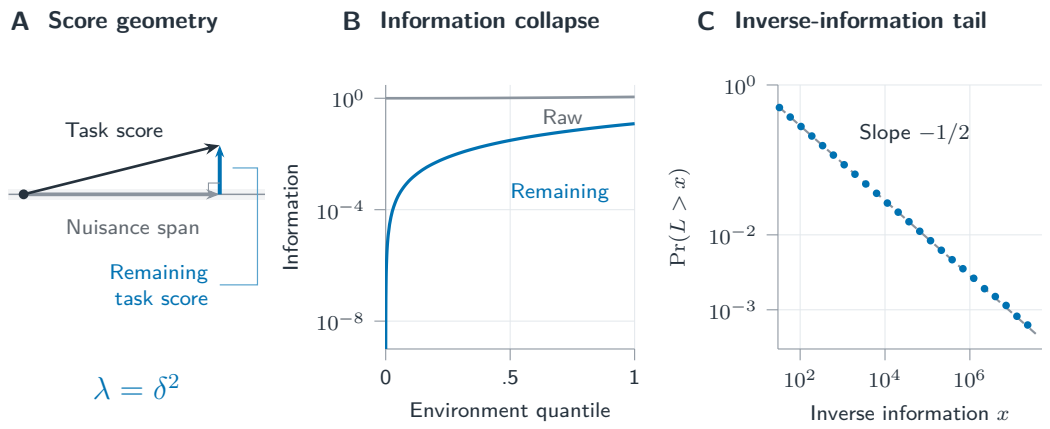


Figure 2: **Synthetic illustration of the geometric tail law in Theorem 1**. (A) The surviving score has norm δ and information $\lambda = \delta^2$. (B) Raw information stays at least one while projected information approaches zero. (C) Empirical inverse-information survival (blue) follows the theoretical $x^{-1/2}$ tail (dashed). B–C share one regenerated seed-11 draw of 300,000 environments, $\delta \sim \mathrm{Uniform}(0, 0.35)$ and $L = (\delta^2 + 10^{-12})^{-1}$. The separately reported five-seed exponent is 0.498 ± 0.003 (nominal 95% normal-approximation interval, ± 1.96 standard errors).

finite precisely when $\beta > 1$. Thus substantial information across most environments can coexist with divergent average inverse information. Appendix A.2 gives the proof and asymptotic constant.

3.2 Efficient Information and Local Classification Difficulty

The geometric tail acquires a classification consequence when nearby nuisance-adjusted alternatives induce quantitatively different optimal decisions. Define classification risk $R_P(g) = P\{g(X) \neq Y\}$, Bayes risk $R_P^\star = \inf_g R_P(g)$, and excess risk $\mathcal{E}_P(g) = R_P(g) - R_P^\star$. For a specified local family, let $\mathfrak{R}_e(n)$ denote minimax expected excess classification risk based on n independent observations from environment e.

Theorem 2 (Local classification difficulty)**.** *Fix $\lambda_e > 0$ and a unit weakest efficient-information direction u_e. For every $0 < t \leq t_{0,e}$, suppose the local family contains $P_{e,+t}, P_{e,-t}$ with task coordinates $\pm t u_e$ and nuisance coordinates allowed to vary. Assume, for constants $K_e, c_e, t_{0,e} > 0$ and $q > 0$,*

$$\mathrm{KL}(P_{e,+t} \| P_{e,-t}) \leq K_e t^2 \lambda_e, \qquad \inf_g \{\mathcal{E}_{P_{e,+t}}(g) + \mathcal{E}_{P_{e,-t}}(g)\} \geq c_e t^q. \tag{5}$$

Then, for a universal constant $c_0 > 0$,

$$\mathfrak{R}_e(n) \geq c_0 c_e \min\{t_{0,e}^q, (n K_e \lambda_e)^{-q/2}\}. \tag{6}$$

The KL condition limits distinguishability after nuisance adjustment; decision separation specifies the cost of confusing the alternatives. Both must hold over the environment-specific radius $t_{0,e}$. For quadratic separation, $q = 2$, when $n K_e \lambda_e \geq t_{0,e}^{-2}$, the bound scales as $c_e/(n K_e \lambda_e)$. The minimum caps the bound at the scale set by the local neighborhood. This is a conditional local result: efficient

information sets the testing scale, while Bayes-decision separation supplies the classification lower bound. The testing proof and tail-aggregation implication are in Appendix A.3.

3.3 The Fisher-Glass Certificate

Upper-tail conditional value-at-risk, CVaR_α, averages the largest $1-\alpha$ fraction of a quantity, with fractional weighting at atoms [Rockafellar and Uryasev, 2000, 2002]. We apply the Fisher-glass certificate [Mansouri and Arai, 2026] to classification environments $e \sim \Pi$, using the weakest task information λ_e and stabilization ε:

$$L_e^{(\varepsilon)} = (\lambda_e + \varepsilon)^{-1}, \qquad I_\alpha^{\mathrm{FG}} = \left[\mathrm{CVaR}_{\alpha, e\sim\Pi}(L_e^{(\varepsilon)})\right]^{-1}, \quad 0 < \alpha < 1. \tag{7}$$

A larger certificate means a smaller inverse-information tail, expressed in information units. Stabilization $\varepsilon > 0$ caps inverse information at $1/\varepsilon$; it does not remove the underlying null geometry. Under Theorem 1's power-law regime, the limit $\varepsilon \downarrow 0$ is positive for $\beta > 1$ and vanishes for $\beta \le 1$.

Proposition 1 (Information coverage)**.** *For $\rho > \varepsilon$,*

$$I_\alpha^{\mathrm{FG}} \ge \rho \quad \Longrightarrow \quad \Pr_{e\sim\Pi}(\lambda_e \ge \rho - \varepsilon) \ge \alpha. \tag{8}$$

The certificate therefore establishes an absolute information floor on at least an α-fraction of the specified environment distribution. Its connection to classification retains the local-radius and decision-separation factors in the preceding bound. The CVaR representation, coverage proof, and limiting interpretation appear in Appendix A.4. Section 4 constructs operational certificates from declared score or feature matrices; Appendix B.1 states their statistical interpretations.

4 Computing and Auditing Fisher-Glass

An operational audit fixes the task contrast, nuisance hypothesis, matrix construction, and audit distribution before measuring information on independent data. Information concerns the declared low-dimensional task and nuisance coordinates. Each experiment specifies coordinate normalization; fitted nuisance probes and temperature calibration are frozen before final evaluation.

4.1 Operational Scores and Nuisance Coordinates

The audit uses a declared matrix target: predictive Fisher under the fitted predictive distribution, observed-label score moments, or feature moments. For binary constructions, write $v_i = (z_i, b_i)^\top$ for the task and nuisance coordinates. Feature moments use $M_i = v_i v_i^\top$; predictive Fisher uses $M_i = p_i(1-p_i)v_i v_i^\top$, where p_i is the model's predicted class probability. The prospective Waterbirds experiment uses the uncentered feature–Schur fraction of fixed task-logit and background-probe coordinates. Hidden-slice discovery uses temperature-calibrated predictive Fisher with shrinkage toward the same-class pooled matrix, $\eta = 0.10$, during search and held-out evaluation; feature moments supply a comparison. Appendix B.1 specifies the constructions, operational subspaces, and computational cost.

The experiments prespecify nuisance coordinates from metadata, fitted annotation probes, or a fixed semantic direction. Without nuisance annotations, that direction is the audited nuisance hypothesis. We report fitting splits, normalization, and nuisance-block conditioning. Appendix C describes gate stability; Appendix D.5 records the predefined-group specification checks.

4.2 Ridge Estimation and Lower Information Bounds

Let M_i be the per-example task–nuisance matrix under the chosen construction, with task and nuisance blocks indexed by T and N. For a predefined slice g of size n_g, form $\widehat{J}_g = n_g^{-1}\sum_{i\in g} M_i$ and compute the ridge Schur estimate

$$\widehat{F}_g^{\mathrm{rid}} = \widehat{J}_{TT,g} - \widehat{J}_{TN,g}(\widehat{J}_{NN,g} + \lambda_{\mathrm{rid}} I)^{-1}\widehat{J}_{NT,g}. \tag{9}$$

Real-data computations use $\lambda_{\mathrm{rid}} = 10^{-3}$. Ridge stabilizes residualization but raises information relative to the unregularized target. Let a_g bound estimation error and d_g bound the ridge-to-

Algorithm 1 Fisher-Glass audit

1: **Specify:** predictor, matrix, coordinates, normalization, calibration, and audit weights.
2: Fix predefined slices, or fit and freeze a constrained gate on discovery data.
3: **Audit independently:** check support and conditioning; compute ridge estimates, unregularized estimates, and lower confidence bounds.
4: **Report:** raw and projected information, surviving fraction, slice support, tail certificate, and uncertainty.

unregularized gap on a simultaneous event over the audited slices. For declared weights Π_G, use

$$\begin{aligned} \underline{\lambda}_g &= [\lambda_{\min}(\widehat{F}_g^{\mathrm{rid}}) - a_g - d_g]_+, \\ \underline{I}_{\alpha,G} &= \left[\mathrm{CVaR}_{\alpha,g\sim\Pi_G}\left((\underline{\lambda}_g + \varepsilon)^{-1}\right)\right]^{-1}. \end{aligned} \tag{10}$$

Here $[x]_+ = \max\{x, 0\}$. On that event, this lower-bounds the unregularized population certificate for the declared matrix and audit distribution. Certification also requires a positive lower confidence bound for $\lambda_{\min}(J_{NN,g})$. The scalar audits use Bonferroni-adjusted Gaussian/Wald moment simulation at nominal 95% confidence, holding the fitted model, evaluated slices, and coordinate-standardization constants fixed. Appendix B records the gap identity, error allowances, and numerical comparison.

4.3 Hidden-Slice Discovery and Held-Out Certification

A gate $h_\phi(x, y) \in [0, 1]$ assigns membership weights. Discovery maximizes inverse projected information subject to effective-sample-size, maximum-weight, and class-mass constraints. We normalize memberships within class, cap excessive weights, and mix with uniform weights as needed to satisfy the effective-sample-size constraint. Five validation folds assess transfer and stability; a final gate is fitted on all discovery data, frozen, and evaluated on an untouched split. The support correction is recomputed on evaluation data under the frozen gate. Its unshrunk target is the weighted ratio $J_\phi = \mathbb{E}[h_\phi M]/\mathbb{E}[h_\phi]$, with positive membership mass. Uncertainty must account for the numerator, membership normalization, and adjustments used to enforce the weight constraints; the Waterbirds target retains the stated shrinkage. Appendix C specifies the constraints, objective, and held-out target.

5 Evaluating Information Audits

We evaluate the surviving fraction under prospective nuisance interventions, inverse-information tails in synthetic experiments, and predictive projected Fisher in held-out discovery.

5.1 Prospective Diagnosis Under a Nuisance Intervention

On Waterbirds [Sagawa et al., 2020], the four groups cross bird class with background type and are written *class/background*. We fully fine-tune ImageNet-initialized MobileNetV3-small models [Howard et al., 2019] with empirical risk minimization. Validation data fit the temperature and cross-fitted background probe. Eligible slices have accuracy at least the overall validation accuracy and negative log-likelihood at most the median across slices. Among them, we select the smallest surviving fraction, $\lambda_{\min}(\widehat{F}_g^{\mathrm{rid}})/\widehat{J}_{TT,g}$. After pilot experiments, this rule is fixed for five additional training seeds, 22–26. Each checkpoint, temperature, probe, selected slice, and audit constant is frozen before intervention scoring.

The rule selects **waterbird/water in all five seeds**: its validation surviving fraction is 24.8%, compared with 59.0% for landbird/land, the other eligible slice. Held-out interventions recompose birds onto same-type or opposite-type backgrounds using the same extracted foreground layer, matte, and task labels. Same-type recomposition controls for the image-construction procedure. Selected-slice accuracy is 92.0% on original images, 93.2% under the control, and 40.6% under opposite-type replacement: a 52.6 ± 1.4 **percentage-point controlled drop**, with a nominal 95% normal-approximation interval across seeds.

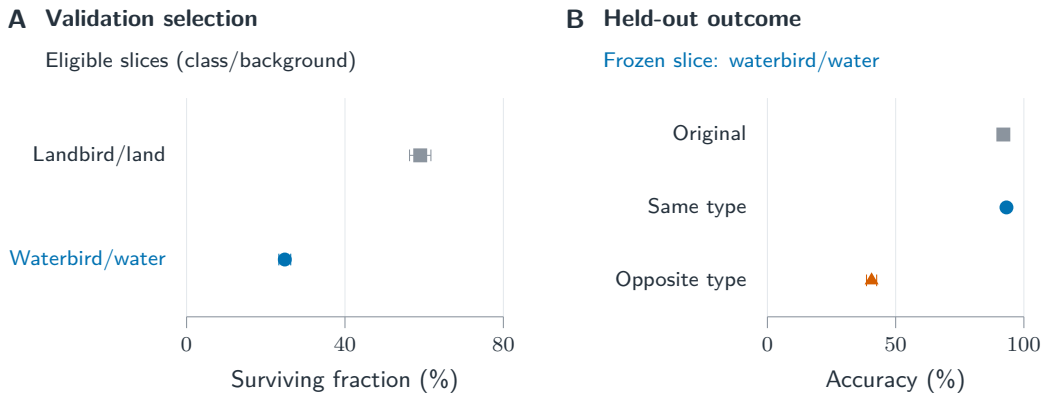


Figure 3: **Validation selection anticipates held-out background sensitivity.** (A) Waterbird/water has the lower uncentered feature–Schur fraction among eligible slices and is selected in all five seeds. Matched minimum-accuracy and maximum-NLL selectors choose the same group in all five. Eligibility and selection are evaluated separately within each seed; names denote class/background. (B) With this selection frozen, ERM accuracy is high on original images and same-type recompositions but falls under opposite-type background recomposition, a controlled difference of 52.6 percentage points. Intervals are nominal 95% normal approximations (± 1.96 standard errors) across seeds 22–26; the same-type arm controls for recomposition.

Within the eligible set, minimum surviving fraction, minimum validation accuracy, and maximum validation NLL select waterbird/water in all five seeds, with a 52.6-point controlled accuracy drop (Figure 3). Minimum raw and absolute projected information select landbird/land, with a 26.7-point drop. Centering switches selection in all five seeds, identifying slice means as a ranking contributor; predictive Fisher retains waterbird/water in all five. Appendix D.2 reports matched selectors, matrix and aggregation comparisons, and individual flip prediction.

5.2 Synthetic Classification and Tail Checks

Draw independent $s, r \sim N(0,1)$, set $z = s + \delta_e r$ and $Y = \mathbf{1}\{z > 0\}$, and evaluate the nuisance-only logit $4s$. Environments use $\delta_e = 0.35 U_e^{1/\gamma}$, $U_e \sim \mathrm{Uniform}(0,1)$, with $\gamma = 1$ in the main experiment. Five-seed mean accuracy is 94.54% and $\mathrm{CVaR}_{0.95}$ of environment-mean NLL is 0.262. Yet $I^{\mathrm{FG}}_{0.95} = (1.22 \pm 0.53) \times 10^{-8}$ with $\varepsilon = 10^{-12}$, despite raw information bounded below by one.

The inverse-information exponent is 0.498 ± 0.003, matching $\beta = \gamma/2 = 0.5$ in the geometric tail law of Theorem 1 (Figure 2). Log-survival regression uses the largest 5% of inverse-information observations. Independent-branch experiments check the scalar portfolio law $\beta = K/2$ (Appendix A.2); reserve experiments check the information floor in Appendix E.1. Uncertainties are nominal 95% normal-approximation intervals across five seeds. These checks exhibit favorable prediction alongside weak projected information and the predicted tail behavior (Appendix D.3).

5.3 Discovering Low-Information Subpopulations

Discovery uses task labels, frozen ImageNet MobileNetV3-small representations for the gate, and a nuisance score obtained by projecting CLIP image embeddings onto a fixed water-minus-land text direction [Radford et al., 2021]. Background and group annotations remain sealed through test evaluation. We search each task class separately with a class-specific logistic gate and select the class–gate pair with highest discovery utility. Gates target 10% within-class mass under Section 4.3's support constraints. Five validation folds assess transfer; the gate is then refit on validation data, frozen, and evaluated once on the untouched test split.

The discovered slice has 98.8% **membership-weighted accuracy** and lower estimated projected predictive Fisher information than high-loss and random selectors targeting the same mass in every seed (Figure 4). Its calibrated predictive information is 0.005 ± 0.001, and its uncalibrated NLL is 0.031 ± 0.017, with 95% Student-t intervals across five paired seeds. Accuracy and NLL use

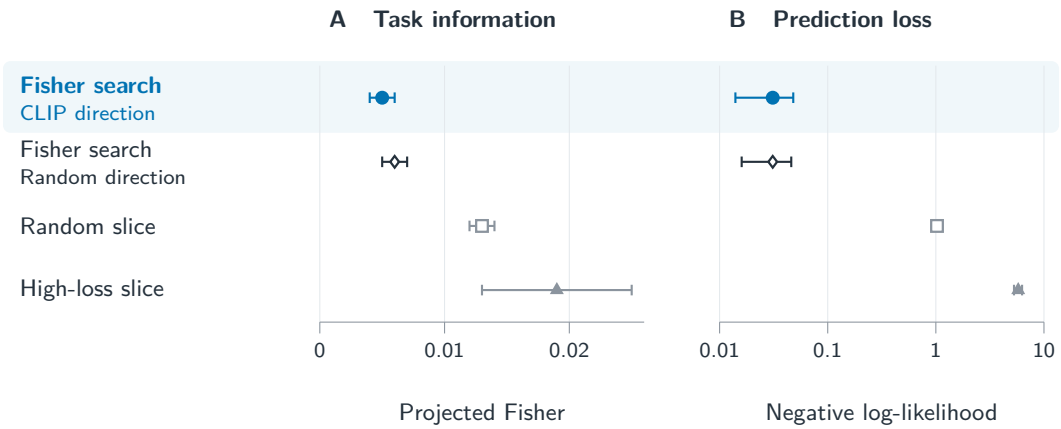


Figure 4: **Discovery finds accurate populations with low predictive information.** Held-out predictive projected Fisher (A) and uncalibrated NLL (B; log scale), with means and 95% Student-t intervals across five paired seeds. The highlighted slice has 98.8% membership-weighted accuracy. Discovery uses task labels and a fixed nuisance direction; group annotations remain sealed and the gate is frozen before test evaluation. Predictive moments retain calibration and same-class pooling with shrinkage 0.10. The random-direction control attains similar information.

support-corrected membership weights without information pooling (Appendix C.1). Its top-10% Jaccard overlap with high-loss selection is 0.001, identifying an accurate population largely distinct from loss-based selection.

The random-direction control attains similar predictive information, with the paired CLIP-minus-random interval spanning zero. The paired intervals for the uncentered and centered feature-moment comparisons with high-loss selection also include zero. These controls reevaluate gates selected using predictive Fisher. The strongest separation concerns predictive information; the incremental contribution of nuisance semantics remains unresolved. Appendix D.4 gives paired intervals, support diagnostics, and revealed composition.

6 Information-Guided Interventions

The audit guides data allocation, training, and acquisition. An *anchor* is an additional observation source with a lower bound on projected task information. When each observation's source is observed, assigning fraction r to an anchor with unregularized information at least $\rho_0 I$ establishes the floor $r\rho_0$, under common coordinates and positive-definite nuisance blocks. This instantiates the Fisher-reserve principle [Mansouri and Arai, 2026] (Appendix E.1). For predictor parameters w, empirical task-risk objective $\widehat{R}_{\mathrm{base}}$ (ERM or group distributionally robust optimization, Group DRO), and penalty coefficient λ_{FG}, training uses equally weighted groups in the inverse-information tail:

$$\min_{w} \ \widehat{R}_{\mathrm{base}}(w) + \lambda_{\mathrm{FG}} \, \mathrm{CVaR}_{\alpha,g}\left[\left\{\lambda_{\min}(\widehat{F}_g^{\mathrm{rid}}(w)) + \varepsilon\right\}^{-1}\right]. \tag{11}$$

Coordinates follow Section 4. Frozen-feature Waterbirds uses validation group labels to select the largest validation certificate among coefficients within two percentage points of the best validation worst-group accuracy.

Fisher-Glass approximately doubles ERM's certificate; with Group DRO it raises both the certificate and mean worst-group accuracy (Table 1). Stronger penalties can reduce accuracy (Appendix E.3), requiring task and validation criteria for selection.

In the prospective fine-tuning experiment, the validation certificate increases from 0.289 to 0.330, while the controlled background-swap drop is 52.9 points versus 52.6 for ERM. In a separate frozen-feature experiment, constrained repair with the Fisher penalty selects the unchanged model in all five seeds. A separate post hoc coefficient-selection analysis increases information in the discovered slice but reduces worst-group accuracy by 12.6 points; its coefficients coincide with validation-NLL minimization in every seed (Appendix E.3).

Table 1: Frozen-feature Waterbirds training. Means with nominal 95% normal-approximation intervals (± 1.96 standard errors) across five seeds; uncentered feature–Schur ridge certificates use $\alpha = 0.95$. Validation selection follows Section 6.

Objective	Average accuracy (%)	Worst-group accuracy (%)	Fisher-Glass certificate
ERM	77.2 ± 0.1	55.7 ± 0.3	0.198 ± 0.001
ERM + Fisher-Glass	80.4 ± 0.2	56.3 ± 0.3	0.399 ± 0.006
Group DRO	80.9 ± 0.3	67.3 ± 0.3	0.233 ± 0.003
Group DRO + Fisher-Glass	80.6 ± 0.9	69.4 ± 1.3	0.614 ± 0.059

For acquisition, candidate a has modified feature/input x_a^{cf}, assigned label y_a, and current predictive loss $\ell_w(x_a^{\mathrm{cf}}, y_a)$. Let $G_\alpha(a)$ be the first-order decrease in inverse-information CVaR along a's specified allocation path. *Tail-transverse influence* is $\mathrm{TTI}(a) = G_\alpha(a)\ell_w(x_a^{\mathrm{cf}}, y_a)$. Gain differentiates the Schur complement along its weakest task direction, weighting environments by tail sensitivity (Appendix E.4). Waterbirds implements $\widetilde{\mathrm{TTI}}_\alpha(a) = \widetilde{G}_\alpha(a)\ell_w(x_a^{\mathrm{cf}}, y_a)$ using a normalized residual-score approximation $\widetilde{G}_\alpha$ (Appendix E.5).

Waterbirds candidates translate features by the difference between the two background-group mean representations within the same bird class, scaled by γ_{cf}. Training data supply means and the nuisance probe; validation data supply candidates, selected features enter retraining, and evaluation is held out. Budgets count acquisitions relative to original training-set size. At $\gamma_{\mathrm{cf}} = 1$ and a 1% budget, $\widetilde{\mathrm{TTI}}_\alpha$ yields a mean worst-group accuracy gain of 4.52 points versus 3.27 for high-loss selection; high loss produces the larger mean certificate gain. At this strength, $\widetilde{\mathrm{TTI}}_\alpha$ has the larger mean worst-group accuracy gain at all three budgets; selection by $\widetilde{G}_\alpha$ alone yields smaller improvements (Appendix E.5).

7 Related Work

Efficient information and classification. Efficient-score theory defines task information after projection away from nuisance tangent spaces [Bickel et al., 1993, Newey, 1990, Tsiatis, 2006]. Fisher-glass theory connects nuisance-adjusted information to inverse-information tails in metrology [Mansouri and Arai, 2026]. Our analysis develops classification bounds and statistical guarantees for population audits. The classification argument uses local testing reductions [Le Cam, 1986, Tsybakov, 2009] under an explicit Bayes-decision separation condition. The operational analysis addresses Schur estimation, ridge effects, and independent evaluation of learned population gates.

Tail risk, robust learning, and gradient geometry. CVaR concentrates objectives on adverse outcomes [Rockafellar and Uryasev, 2000, 2002]; Group DRO applies robust optimization to predictive loss across groups [Sagawa et al., 2020]. Invariant Risk Minimization seeks shared optimal classifiers [Arjovsky et al., 2019], while Fishr regularizes domain-level gradient variances [Rame et al., 2022]. We use inverse projected-information tail risk as a training penalty alongside empirical risk or Group DRO, and evaluate information and predictive performance together.

Hidden stratification and slice discovery. Hidden-stratification research motivates finer population evaluation [Oakden-Rayner et al., 2020]. GEORGE clusters representations to infer subclasses [Sohoni et al., 2020], JTT upweights initially misclassified examples [Liu et al., 2021], and AGRO learns adversarial group assignments using prediction loss [Paranjape et al., 2023]. Our search targets low predictive projected information given task labels and a fixed nuisance direction. Support constraints and an independently evaluated frozen gate make accurately classified, low-loss populations eligible audit targets.

Information-based design and acquisition. E-optimal design emphasizes the smallest information eigenvalue [Pukelsheim, 2006]; BAIT uses Fisher information for neural active learning [Ash et al., 2021]. TTI measures local information gain along a weak nuisance-adjusted task direction, weights environments by tail sensitivity, and multiplies by current task loss. Acquisition experiments assess realized information and accuracy changes after retraining.

8 Limitations

Specification and attribution. The audit depends on task contrast, nuisance space, and representation; nuisance variation must preserve the task. Absolute eigenvalues depend on task scaling; ridge estimates also on nuisance scaling. Standardization does not establish representation invariance. Ridge inflates projected information and can conceal an unregularized collapse; a small ridge parameter controls this bias only when the nuisance block has a positive conditioning margin. Predictive Fisher weights confidence; unweighted Schur statistics describe residual feature geometry. Their relation to population efficient information depends on the score model. Centering reverses the prospective ranking in all five seeds, so its original uncentered diagnostic depends partly on slice means. Comparable CLIP and random-direction discovery results and unweighted-control intervals containing zero leave semantic nuisance projection's incremental contribution unresolved. Confidence-matched comparisons and systematic nuisance controls remain necessary; local certificates do not guarantee reliability outside audited populations.

Local theory and audit scope. Efficient information, separation strength, and local radius jointly determine classification bounds, which require regularity and quantitative Bayes-decision separation. Sample size counts the audited environment's observations. Latent-environment pooling also depends on prevalence, shared information, and discovery. Coverage is relative to the declared environment distribution. Finite audits cover included groups or frozen slice rules; small mass and weak nuisance conditioning amplify uncertainty. Bonferroni-adjusted Gaussian/Wald coverage is nominal, conditional on fitted models, evaluated families, and realized coordinate-standardization constants; uncertainty in evaluation-fitted constants is omitted. Repeated searches must account for certification-data reuse. Empirical sample-size scaling and achievable classification rates remain open.

Development history and empirical breadth. The prospective rule was frozen after pilot experiments for fresh seeds on the same benchmark. Independent confirmatory datasets and matched evaluations of comparative selection gains are needed. The probe's background attribute defines the intervention; same-type recomposition controls shared construction effects. Evidence centers on MobileNetV3-small image classifiers, supplementary CivilComments TF-IDF diagnostics (Appendix D.5), and synthetic constructions. Transformer, segmentation, and diffusion applications, including nuisance discovery without metadata or prior structure, require further evaluation. The unregularized training audits concern the fresh models in Appendix B.4.

Information, performance, and intervention scope. Information gains need not repair predictions: prospective swap sensitivity persists, constrained slice repair selects unchanged models, and post hoc selection increases slice information while reducing worst-group accuracy. Prospective FG selection required fallbacks in seeds 22 and 24 because no candidate met ERM's overall validation accuracy minus one percentage point. Post hoc choices coincide with validation-NLL minimization. The proposed threshold-based selection policy requires prospective validation against the deployment objective. Reserves assume an informative anchor and stated information-accumulation design; a general admission procedure for nuisance relevance, task preservation, split stability, and conditioning remains to be evaluated. TTI measures local predictor sensitivity; finite interventions and retraining are empirical tests. Transfer of annotation-conditioned feature shifts to realizable label-preserving interventions and other candidate families remains unestablished.

9 Conclusion

We develop classification and auditing methods for evaluating nuisance-projected task information in trained predictors. A local excess-risk lower bound connects efficient information to learning difficulty under explicit decision-separation conditions. Regularization-aware bounds and independent evaluation support audits of predefined groups and learned subpopulations. On Waterbirds, a validation-selected group exhibits a 52.6-point controlled accuracy drop under background replacement, while held-out discovery identifies accurate populations with low estimated projected predictive Fisher information. Training and acquisition experiments characterize the joint response of information and predictive performance. These results support evaluating classifiers through observed errors, specified information targets, and sensitivity to declared nuisance changes.

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A Proofs of the Core Results

Throughout, task coordinates use the common normalization of Section 2, and environments follow the declared distribution Π.

A.1 Projection, Normalization, and Task–Nuisance Tangency

Use the centered, square-integrable scores and blocks $J_{AB,e}$ of Section 2.1. Suppose the d_N coordinates of the finite nuisance-score vector span $\mathcal{T}_{N,e}$ and $J_{NN,e} \succ 0$. With $\Gamma_e = J_{TN,e}J_{NN,e}^{-1}$,

$$\begin{aligned} \mathbb{E}_{P_e}[(s_{T,e} - \Gamma_e s_{N,e})s_{N,e}^\top] &= J_{TN,e} - \Gamma_e J_{NN,e} = 0, \\ F_e = \mathbb{E}_{P_e}[s_{T,e}^{\text{eff}}(s_{T,e}^{\text{eff}})^\top] &= J_{TT,e} - J_{TN,e}J_{NN,e}^{-1}J_{NT,e}. \end{aligned} \tag{A.1}$$

Because $\Gamma_e s_{N,e}$ lies componentwise in the nuisance span, orthogonality identifies the residual as the efficient score in Equation (1) [Bickel et al., 1993, Tsiatis, 2006]. Equivalently,

$$u^\top F_e u = \min_{v \in \mathbb{R}^{d_N}} \mathbb{E}_{P_e}[(u^\top s_{T,e} - v^\top s_{N,e})^2], \qquad v^\star = J_{NN,e}^{-1}J_{NT,e}u. \tag{A.2}$$

Hence $F_e \succeq 0$, with $u^\top F_e u = 0$ exactly when $u^\top s_{T,e}$ belongs to the nuisance span. The projection definition extends directly to the closed span of an infinite-dimensional nuisance tangent space.

For separate invertible transformations $\widetilde{s}_T = Us_T$ and $\widetilde{s}_N = Vs_N$, the nuisance span is unchanged and

$$\widetilde{s}_T^{\text{eff}} = Us_T^{\text{eff}}, \qquad \widetilde{F} = UFU^\top; \qquad \frac{F}{J_{TT}} = 1 - \rho^2 \quad \text{in the scalar case.} \tag{A.3}$$

Here $\rho = J_{TN}/\sqrt{J_{TT}J_{NN}}$ requires positive raw task and nuisance information. Exact information is invariant to nuisance-basis changes; task congruence preserves rank but generally changes eigenvalues. The scalar surviving fraction is invariant to nonzero task rescaling and invertible nuisance-basis changes. Nuisance ridge preserves the latter invariance only if $\lambda_{\text{rid}}I$ becomes $\lambda_{\text{rid}}VV^\top$ under $s_N \mapsto Vs_N$ (Appendix B).

For the tangency construction in Section 2.2, let $b_e, r_e \in L_0^2(P_e)$ be orthogonal unit vectors, $\mathcal{T}_{N,e} = \text{span}\{b_e\}$, and $a_e = b_e + \delta_e r_e$ with $\delta_e \geq 0$. The projection coefficient is $\langle a_e, b_e\rangle_e/\|b_e\|_e^2 = 1$, so

$$a_e^{\text{eff}} = \delta_e r_e, \qquad F_e = \|\delta_e r_e\|_e^2 = \delta_e^2, \qquad J_{TT,e} = 1 + \delta_e^2, \qquad \frac{F_e}{J_{TT,e}} = \frac{\delta_e^2}{1+\delta_e^2}. \tag{A.4}$$

This establishes Equation (3); coexistence with low prediction risk additionally requires a successful nuisance-based predictor.

A.2 Normal Coordinates and the Inverse-Information Tail

Under Theorem 1, the zero set $\mathcal{Z} = \{\lambda = 0\}$ is a nonempty compact C^2 embedded manifold without boundary, of codimension $c \in \{1, \ldots, d\}$. Write $U_r(\mathcal{Z}) = \{z : \text{dist}(z, \mathcal{Z}) < r\}$ for its tube, $N_s\mathcal{Z}$ for the normal space at s, $S(N_s\mathcal{Z})$ for its unit sphere, and $\text{vol}_{\mathcal{Z}}$ for induced volume. The nonnegative λ is continuous on $\overline{U}_{r_\star}(\mathcal{Z})$ and bounded away from zero outside it. In coordinates $z = s + r\omega$, assume uniformly over $\omega \in S(N_s\mathcal{Z})$ and $s \in \mathcal{Z}$

$$\lambda(s + r\omega) = a(s,\omega)r^{2m}(1 + o(1)), \qquad p(s + r\omega) = b(s,\omega)r^\nu(1 + o(1)), \tag{A.5}$$

where p is the Lebesgue density, $m > 0$, $\nu > -c$, and continuous a, b have positive uniform lower and upper bounds. Put

$$\beta = \frac{c+\nu}{2m}, \qquad C_{\mathcal{Z}} = \frac{1}{c+\nu}\int_{\mathcal{Z}}\int_{S(N_s\mathcal{Z})} b(s,\omega)a(s,\omega)^{-\beta}\, d\sigma_s(\omega)\, d\,\text{vol}_{\mathcal{Z}}(s). \tag{A.6}$$

The sphere measure σ_s is unnormalized; for $c = 1$, each of the two normal directions has mass one.

Proof of Theorem 1. For any $0 < r_0 < r_\star$, continuity and absence of zeros on the compact annulus $r_0 \leq \text{dist}(z, \mathcal{Z}) \leq r_\star$ give a positive minimum of λ. Together with the bound outside the outer tube, this localizes $\{\lambda < t\}$ to $U_{r_0}(\mathcal{Z})$ for all sufficiently small t. The tubular volume element is

$dz = j(s, r\omega) r^{c-1}\, dr\, d\sigma_s\, d\,\mathrm{vol}_{\mathcal{Z}}$, where the coordinate Jacobian satisfies $j = 1 + O(r)$ uniformly. Thus, for any $0 < \zeta < 1$, uniformity permits r_0 small enough that

$$\begin{aligned} (1-\zeta) a r^{2m} &\le \lambda(s + r\omega) \le (1+\zeta) a r^{2m}, \\ (1-\zeta) b r^{\nu} &\le p(s + r\omega) j(s, r\omega) \le (1+\zeta) b r^{\nu} \end{aligned} \qquad (0 < r < r_0). \tag{A.7}$$

For small t, all radial thresholds lie inside this tube. The event contains $r < (t/((1+\zeta)a))^{1/(2m)}$ and is contained in $r < (t/((1-\zeta)a))^{1/(2m)}$. Integrating uses $\int_0^{(t/a)^{1/(2m)}} b r^{c+\nu-1}\, dr = b a^{-\beta} t^{\beta}/(c+\nu)$ and gives

$$(1-\zeta)(1+\zeta)^{-\beta} C_{\mathcal{Z}} \le \frac{\Pr(\lambda < t)}{t^{\beta}} \le (1+\zeta)(1-\zeta)^{-\beta} C_{\mathcal{Z}}. \tag{A.8}$$

Let $t \downarrow 0$ and then $\zeta \downarrow 0$. Compactness and the coefficient bounds ensure $0 < C_{\mathcal{Z}} < \infty$, so $\Pr(\lambda < t) \sim C_{\mathcal{Z}} t^{\beta}$. Since $c \ge 1$, $\mathcal{Z}$ has zero Lebesgue and environment measure; substituting $t = x^{-1}$ proves the inverse tail in the next display. □

The tail-integral identity then yields, for every $p_0 > 0$,

$$\begin{aligned} &\Pr(\lambda^{-1} > x) \sim C_{\mathcal{Z}} x^{-\beta}, \\ &\mathbb{E}[\lambda^{-p_0}] = p_0 \int_0^{\infty} x^{p_0 - 1} \Pr(\lambda^{-1} > x)\, dx < \infty \quad \Longleftrightarrow \quad p_0 < \beta. \end{aligned} \tag{A.9}$$

The bounded-x part is finite; the tail has the convergence criterion of $\int^{\infty} x^{p_0 - \beta - 1}\, dx$. In particular, the inverse mean is finite exactly for $\beta > 1$, with logarithmic divergence at $\beta = 1$.

Scalar branches. For $\lambda = \delta^2$, $\Pr(\delta < t) \sim C_\delta t^{\gamma}$ gives exponent $\gamma/2$. More generally, let $\lambda_K = \sum_{k=1}^{K} w_k \delta_k^2$, $w_k > 0$, where $\delta \in [0, \infty)^K$ has joint density f continuous and positive at the origin from within the orthant. Changing variables gives the scalar portfolio law [Mansouri and Arai, 2026]:

$$\begin{aligned} \Pr(\lambda_K < t) &= \frac{t^{K/2}}{\sqrt{\prod_k w_k}} \int_{B_K^+} f\big(\sqrt{t/w_1} u_1, \ldots, \sqrt{t/w_K} u_K\big)\, du \\ &\sim \frac{f(0) v_K}{2^K \sqrt{\prod_k w_k}}\, t^{K/2}. \end{aligned} \tag{A.10}$$

Here B_K^+ is the positive-orthant unit ball and v_K the full ball's volume. Independence gives $f(0) = \prod_k f_k(0+)$ for branch marginal densities f_k; the inverse mean is finite for $K \ge 3$. This calculation concerns additive scalar information or a fixed task direction. A minimum-eigenvalue conclusion instead requires applying Theorem 1 to that eigenvalue.

A.3 Local Classification: Testing Reduction and Tail Aggregation

Let $\mathcal{E}_P(g) = R_P(g) - R_P^{\star}$. Fix a local family $\mathcal{P}_e^{\mathrm{loc}}$ independently of sample size, and define

$$\mathfrak{R}_e(n) = \inf_{\widehat{g}_n} \sup_{P \in \mathcal{P}_e^{\mathrm{loc}}} \mathbb{E}_{S_n \sim P^{\otimes n}}[\mathcal{E}_P(\widehat{g}_n(S_n))], \tag{A.11}$$

where learning rules are measurable and n counts independent observations from environment e; independent learner randomization is included in the expectation. Assume $\lambda_e > 0$ and a unit weakest efficient-information direction u_e. For every $0 < t \le t_{0,e}$, the fixed family contains $P_{e,\pm t}$ with task coordinates $\pm t u_e$ and potentially varying nuisance coordinates. For fixed $K_e, c_e, t_{0,e} > 0$ and $q > 0$, assume both bounds in Equation (5) hold throughout that radius.

Proof of Theorem 2. Fix an admissible t and abbreviate $P_{\pm} = P_{e,\pm t}$. For any learning rule, the decision-separation condition $\mathcal{E}_{P_+}(g) + \mathcal{E}_{P_-}(g) \ge c_e t^q$ implies that choosing the alternative with smaller excess risk makes any testing error cost at least $c_e t^q/2$ under the true alternative. Explicitly,

$$\begin{aligned} \psi_t(S_n) = + \iff \mathcal{E}_{P_+}(\widehat{g}_n(S_n)) \le \mathcal{E}_{P_-}(\widehat{g}_n(S_n)), \\ \max_{\sigma \in \{+,-\}} \mathbb{E}_{P_\sigma^{\otimes n}}[\mathcal{E}_{P_\sigma}(\widehat{g}_n)] \ge \frac{c_e t^q}{4} \{P_+^{\otimes n}(\psi_t = -) + P_-^{\otimes n}(\psi_t = +)\}. \end{aligned} \tag{A.12}$$

The standard exponential two-point testing inequality [Tsybakov, 2009, Lemma 2.6], together with product additivity of KL and the assumed KL bound, gives

$$P_+^{\otimes n}(\psi_t = -) + P_-^{\otimes n}(\psi_t = +) \geq \tfrac{1}{2}\exp[-n\,\mathrm{KL}(P_+\|P_-)] \geq \tfrac{1}{2}e^{-nK_e t^2 \lambda_e},$$
$$\mathfrak{R}_e(n) \geq \frac{c_e}{8} t^q e^{-nK_e t^2 \lambda_e}. \tag{A.13}$$

The second line takes the supremum over the fixed family containing both alternatives and then the infimum over learning rules. Choose

$$t_n = \min\{t_{0,e}, (nK_e\lambda_e)^{-1/2}\}, \qquad nK_e\lambda_e t_n^2 \leq 1,$$
$$\mathfrak{R}_e(n) \geq c_0 c_e \min\{t_{0,e}^q, (nK_e\lambda_e)^{-q/2}\}, \qquad c_0 = \frac{e^{-1}}{8}. \tag{A.14}$$

Independent learner randomization has the same law under both alternatives and leaves their KL divergence unchanged, so the argument also covers randomized rules. □

Quadratic decision separation. Suppose the binary alternatives share a feature marginal Q_e, with regression functions $\eta_{\pm,t}(x) = P_{e,\pm t}(Y = 1 \mid X = x)$ and Bayes rules $g^\star_{\pm,t}(x) = \mathbf{1}\{\eta_{\pm,t}(x) \geq 1/2\}$. Let $D_{e,t} = \{g^\star_{+,t} \neq g^\star_{-,t}\}$ be their disagreement set. Applying the binary excess-risk identity [Audibert and Tsybakov, 2007, Eq. (1.1)] gives

$$\inf_g \{\mathcal{E}_{P_{e,+t}}(g) + \mathcal{E}_{P_{e,-t}}(g)\}$$
$$= \int_{D_{e,t}} \min\{|2\eta_{+,t}(x) - 1|, |2\eta_{-,t}(x) - 1|\}\, dQ_e(x). \tag{A.15}$$

Pointwise, a classifier must disagree with at least one Bayes rule on $D_{e,t}$; choosing the smaller margin attains the integral for unrestricted measurable binary classifiers, while it remains a lower bound for a restricted class. If, for every $0 < t \leq t_{0,e}$, a measurable $A_{e,t} \subseteq D_{e,t}$ has $Q_e(A_{e,t}) \geq c_{\mathrm{mass},e}t$ and both margins at least $c_{\mathrm{margin},e}t$, with positive constants independent of t, the integral is at least $c_{\mathrm{mass},e}c_{\mathrm{margin},e}t^2$. Thus $q = 2$ and $c_e = c_{\mathrm{mass},e}c_{\mathrm{margin},e}$ satisfy decision separation; the KL condition is still required separately.

Environmental aggregation. If these assumptions hold for Π-almost every environment and all quantities are measurable, monotonicity and positive homogeneity of CVaR give

$$\mathcal{B}_e(n) = c_e \min\{t_{0,e}^q, (nK_e\lambda_e)^{-q/2}\},$$
$$\mathrm{CVaR}_{\alpha,e\sim\Pi}(\mathfrak{R}_e(n)) \geq c_0\, \mathrm{CVaR}_{\alpha,e\sim\Pi}(\mathcal{B}_e(n)). \tag{A.16}$$

Separation c_e, distinguishability K_e, and radius $t_{0,e}$ remain inside the aggregation. For $q = 2$ and $nK_e\lambda_e \geq t_{0,e}^{-2}$, $\mathcal{B}_e(n) = c_e/(nK_e\lambda_e)$; outside that interior regime the radius truncates the bound.

A.4 CVaR, Information Coverage, and Stabilization

For $0 < \alpha < 1$ and nonnegative Z, the CVaR minimization formula holds for integrable Z [Rockafellar and Uryasev, 2002, Theorem 10]; the threshold-weight argument below gives its dual:

$$\mathrm{CVaR}_\alpha(Z) = \inf_{\tau\in\mathbb{R}} \left\{\tau + \frac{\mathbb{E}[(Z-\tau)_+]}{1-\alpha}\right\} = \sup_{\xi\in\mathcal{W}_\alpha} \mathbb{E}[\xi Z],$$
$$\mathcal{W}_\alpha = \{\xi : 0 \leq \xi \leq (1-\alpha)^{-1},\ \mathbb{E}\xi = 1\}. \tag{A.17}$$

Both values are $+\infty$ if $\mathbb{E}Z = +\infty$. For integrable Z, a maximizing weight equals $(1-\alpha)^{-1}$ above $v_\alpha = \inf\{v : \Pr(Z \leq v) \geq \alpha\}$, zero below it, and fills the remaining mass at the threshold. Such fractional weighting is possible because $\Pr(Z > v_\alpha) \leq 1 - \alpha \leq \Pr(Z \geq v_\alpha)$. The dual gives monotonicity, positive homogeneity, and $v_\alpha \leq \mathrm{CVaR}_\alpha(Z)$. For the finite audit distribution Π_G, let $\pi_g = \Pi_G(\{g\}) \geq 0$, $\sum_g \pi_g = 1$. Then

$$\mathrm{CVaR}_{\alpha,\pi}(Z_g) = \max_{\substack{\sum_g \omega_g = 1 \\ 0\leq\omega_g\leq\pi_g/(1-\alpha)}} \sum_g \omega_g Z_g. \tag{A.18}$$

These normalized tail weights, including threshold atoms, also govern Appendix E.4.

Proof of Proposition 1. Let $\varepsilon > 0$ and $\lambda_e \in [0, \infty)$ be finite almost surely. Write $L_e^{(\varepsilon)} = (\lambda_e + \varepsilon)^{-1}$ and $I_\alpha^{\mathrm{FG}}(\varepsilon)$ for the certificate in Equation (7). If $I_\alpha^{\mathrm{FG}}(\varepsilon) \geq \rho > \varepsilon$, then

$$\begin{aligned} v_\alpha(L^{(\varepsilon)}) \leq \mathrm{CVaR}_\alpha(L^{(\varepsilon)}) = [I_\alpha^{\mathrm{FG}}(\varepsilon)]^{-1} \leq \rho^{-1}, \\ \Pr_{e\sim\Pi}(\lambda_e \geq \rho - \varepsilon) = \Pr_{e\sim\Pi}(L_e^{(\varepsilon)} \leq \rho^{-1}) \geq \alpha. \end{aligned} \tag{A.19}$$

The last step is the quantile definition and includes equality at the information threshold. □

Since $0 < L^{(\varepsilon)} \leq \varepsilon^{-1}$, always $I_\alpha^{\mathrm{FG}}(\varepsilon) \geq \varepsilon$; only $\rho > \varepsilon$ certifies a strictly positive floor. With four equally weighted groups and $\alpha = 0.95$, one group already contains the entire upper-tail mass, so

$$\mathrm{CVaR}_{0.95}(L^{(\varepsilon)}) = \max_g L_g^{(\varepsilon)}, \qquad I_{0.95}^{\mathrm{FG}}(\varepsilon) = \min_g \lambda_g + \varepsilon. \tag{A.20}$$

Exact limit and positivity. Set $L^{(0)} = \lambda^{-1}$, with $1/0 = +\infty$. As $\varepsilon \downarrow 0$, $L^{(\varepsilon)} \uparrow L^{(0)}$, and monotone convergence applies to $\mathbb{E}[\xi L^{(\varepsilon)}]$ for every feasible ξ. The supremum over ξ commutes with the increasing supremum along any sequence $\varepsilon_k \downarrow 0$, giving

$$\begin{aligned} \mathrm{CVaR}_\alpha(L^{(\varepsilon)}) \uparrow \mathrm{CVaR}_\alpha(\lambda^{-1}), \\ I_\alpha^{\mathrm{FG}}(\varepsilon) \downarrow I_\alpha^{\mathrm{FG}}(0) := \left[\mathrm{CVaR}_\alpha(\lambda^{-1})\right]^{-1}, \qquad 1/(+\infty) = 0. \end{aligned} \tag{A.21}$$

For nonnegative Z, the feasible weight $\xi = 1$ and variational choice $\tau = 0$ imply

$$\mathbb{E}Z \leq \mathrm{CVaR}_\alpha(Z) \leq \frac{\mathbb{E}Z}{1-\alpha}, \qquad I_\alpha^{\mathrm{FG}}(0) > 0 \iff \mathbb{E}[\lambda^{-1}] < \infty. \tag{A.22}$$

Under the geometric tail assumptions, this is equivalent to $\beta = (c + \nu)/(2m) > 1$; the exact certificate vanishes for $\beta \leq 1$. Information coverage is a population statement under Π, separate from the confidence level of an estimated lower certificate.

B Estimation and Certification

The bounds condition on a predictor, coordinates, calibration, and slices fixed independently of evaluation data. Evaluation-fitted normalization requires additional error control. Implementation details and numerical records appear in the artifact's `docs/extended_appendix_details/` (see the artifact availability statement).

B.1 Operational Models and Matrix Targets

For evaluation distribution P_g and calibrated K-class logits ℓ_w, maps $\Phi_T(x) \in \mathbb{R}^{K\times d_T}$ and $\Phi_N(x) \in \mathbb{R}^{K\times d_N}$ convert task and nuisance parameters into logit changes:

$$\ell_{\theta,\mu}(x) = \ell_w(x) + \Phi_T(x)\theta + \Phi_N(x)\mu, \qquad D = (\Phi_T\ \Phi_N), \quad d = d_T + d_N. \tag{B.1}$$

With $p = \mathrm{softmax}(\ell_w)$, $H = \mathrm{diag}(p) - pp^\top$, one-hot label vector $\mathbf{e}_Y$, and normalized feature coordinates $v = (v_T^\top\ v_N^\top)^\top$, distinguish

$$\begin{aligned} M^{\mathrm{pred}} = D^\top H D, \qquad M^{\mathrm{obs}} = ss^\top, \quad s = D^\top(\mathbf{e}_Y - p), \\ M^{\mathrm{feat}} = vv^\top, \qquad J_g = \mathbb{E}_{P_g}[M]. \end{aligned} \tag{B.2}$$

For a binary scalar logit $\ell_{\theta,\mu} = \ell_w + \theta z + \mu b$, this gives $M^{\mathrm{pred}} = p(1-p)vv^\top$, with $p = \sigma(\ell_w)$ and $v = (z, b)^\top$. Predictive scores have conditional mean zero under the fitted label distribution even when features do not; their Schur complement is efficient information for this operational model. Observed-label moments equal predictive Fisher under conditional correct specification, $P_g(Y = \cdot \mid X) = p(X)$; otherwise their centered covariance subtracts $\mathbb{E}[s]\mathbb{E}[s]^\top$. Feature geometry satisfies $J_g = \Sigma_g + \mu_g\mu_g^\top$, with $\mu_g = \mathbb{E}[v]$ and $\Sigma_g = \mathrm{Cov}(v)$. The prospective centered comparison uses within-slice covariance Σ_g with divisor n_g and unchanged global scales, removing the mean contribution without rescaling.

Prospective selection uses uncentered feature–Schur fractions; Appendix D.2 compares centered and predictive targets. Within-group normalization of $p(1-p)$ weights changes the effective nuisance ridge and differs from the predictive moment. Hidden discovery leaves this factor unnormalized and shrinks toward the same-class pooled matrix, with $\eta = 0.10$. Removing ridge leaves the declared matrix, including shrinkage, unchanged. Normalization conventions and nuisance-specification checks appear in Appendices D and D.5.

Operational subspaces and cost. A task coordinate specifies the predictive contrast to be audited; nuisance coordinates specify variations expected to preserve it. Candidate constructions include an answer-logit contrast with controlled prompt-template variation for a transformer, a region-level class contrast with validated appearance changes for segmentation, and a conditional denoising-score contrast for a diffusion model. Logits, probes, intermediate representations, and low-rank parameter perturbations provide finite-dimensional coordinates. For total operational dimension $d = d_T + d_N$, dense moment storage costs $O(d^2)$, and direct factorization or eigendecomposition costs $O(d^3)$ per slice. Minibatch accumulation and iterative eigensolvers support larger operational subspaces; feature or score extraction remains dependent on the backbone and input size. These examples are candidate extensions for future evaluation.

Admission of new nuisance specifications. A proposed admission protocol prespecifies independent checks of nuisance relevance, task preservation, split stability, and nuisance-block conditioning. An identifiability conclusion requires all four checks to pass; a failed prerequisite is recorded as an unvalidated specification. For a finite admitted family, report the minimum certificate with uncertainty accounting for the family. The present experiments evaluate prespecified nuisance hypotheses, with implemented checks and sensitivity analyses in Appendices B.4, C, and D.5. The general admission procedure remains to be evaluated.

B.2 Ridge Regularization and the Unregularized Target

For $J = \left(\begin{smallmatrix} A & B \\ B^\top & C \end{smallmatrix}\right) \succeq 0$, with $A = J_{TT}$, $B = J_{TN}$, $C = J_{NN} \succ 0$, and $\lambda_{\mathrm{rid}} > 0$, define

$$S_0(J) = A - BC^{-1}B^\top, \qquad S_{\mathrm{rid}}(J) = A - B(C + \lambda_{\mathrm{rid}} I)^{-1}B^\top. \tag{B.3}$$

Both are positive semidefinite. The resolvent identity, whose inverse factors commute, gives

$$\begin{aligned} \Delta_{\mathrm{rid}}(J) &:= S_{\mathrm{rid}}(J) - S_0(J) = \lambda_{\mathrm{rid}} BC^{-1}(C + \lambda_{\mathrm{rid}} I)^{-1}B^\top \succeq 0, \\ \|\Delta_{\mathrm{rid}}(J)\|_{\mathrm{op}} &\le \frac{\lambda_{\mathrm{rid}} M^2}{\kappa(\kappa + \lambda_{\mathrm{rid}})} \quad \text{if } C \succeq \kappa I,\ \kappa > 0,\ \|B\|_{\mathrm{op}} \le M. \end{aligned} \tag{B.4}$$

Conditioning is essential: scalar blocks $A = 1$, $B = \sqrt{c}$, $C = c > 0$ give a positive-semidefinite joint matrix with $S_0 = 0$ and $S_{\mathrm{rid}} = \lambda_{\mathrm{rid}}/(c + \lambda_{\mathrm{rid}}) \to 1$ as $c/\lambda_{\mathrm{rid}} \downarrow 0$. Comparisons therefore retain the declared standardized nuisance coordinates and ridge value.

B.3 Error Allowances and Lower Certificates

Proposition B.1 (Simultaneous lower certificate)**.** *Fix G groups and weights $\pi_g > 0$ summing to one. Within each group, let M_{gi} be i.i.d. with mean J_g, $0 \preceq M_{gi} \preceq R_g I_d$, for specified finite R_g, and $\widehat{J}_g = n_g^{-1} \sum_i M_{gi}$. For $0 < \delta < 1$, set*

$$\begin{aligned} u_\delta &= \log(2dG/\delta), \qquad t_g = R_g \left(\sqrt{2u_\delta/n_g} + 2u_\delta/(3n_g) \right), \\ \mathcal{E}_\delta &= \bigcap_g \{ \|\widehat{J}_g - J_g\|_{\mathrm{op}} \le t_g \}, \qquad \Pr(\mathcal{E}_\delta) \ge 1 - \delta. \end{aligned} \tag{B.5}$$

Assume each $C_g \succ 0$. Define observable bounds and allowances by

$$\begin{aligned} \overline{M}_g &= \|\widehat{B}_g\|_{\mathrm{op}} + t_g, & \underline{\kappa}_g &= \lambda_{\min}(\widehat{C}_g) - t_g, \\ h_{\mathrm{rid}}(t, M) &= t + (2Mt + t^2)/\lambda_{\mathrm{rid}} + M^2 t/\lambda_{\mathrm{rid}}^2, & a_g &= h_{\mathrm{rid}}(t_g, \overline{M}_g), \\ d_g &= \frac{\lambda_{\mathrm{rid}} \overline{M}_g^2}{\underline{\kappa}_g(\underline{\kappa}_g + \lambda_{\mathrm{rid}})} & & \text{when all } \underline{\kappa}_g > 0. \end{aligned} \tag{B.6}$$

Report the lower certificate in (10) with audit distribution π only when these conditioning margins are positive; otherwise record conditioning failure. Its probability of exceeding the unregularized population certificate is at most δ. On $\mathcal{E}_\delta$, a reported level $\underline{I}_{\alpha,\pi} \ge \rho > \varepsilon$ also implies $\sum_{g:\lambda_{\min}(S_0(J_g)) \ge \rho - \varepsilon} \pi_g \ge \alpha$.

Proof. For $X_{gi} = M_{gi} - J_g$, $\|X_{gi}\|_{\mathrm{op}} \le R_g$ and $\mathbb{E}[X_{gi}^2] = \mathbb{E}[M_{gi}^2] - J_g^2 \preceq R_g^2 I$. Matrix Bernstein [Tropp, 2012, Theorem 1.4] with the displayed t_g gives failure at most δ/G per group; the union bound allows dependence between group estimates. On this event, block extraction and Weyl give $\|B_g\|_{\mathrm{op}} \le \overline{M}_g$ and $C_g \succeq \underline{\kappa}_g I$, so (B.4) bounds the ridge gap by d_g.

For the estimation error, write $Q = (C + \lambda_{\mathrm{rid}} I)^{-1}$, $\widehat{Q} = (\widehat{C} + \lambda_{\mathrm{rid}} I)^{-1}$, and $\Delta B = \widehat{B} - B$. Positive semidefiniteness and the resolvent identity give $\|Q\|, \|\widehat{Q}\| \le \lambda_{\mathrm{rid}}^{-1}$ and $\|\widehat{Q} - Q\| \le t/\lambda_{\mathrm{rid}}^2$. Decompose the quadratic-block difference as $\Delta B \widehat{Q} \widehat{B}^\top + B \widehat{Q} \Delta B^\top + B(\widehat{Q} - Q) B^\top$. If $\|B\| \le M$, its three norms are bounded by $t(M + t)/\lambda_{\mathrm{rid}}$, $Mt/\lambda_{\mathrm{rid}}$, and $M^2 t/\lambda_{\mathrm{rid}}^2$. Adding the task-block error t proves the bound $h_{\mathrm{rid}}(t, M)$ and hence a_g.

Writing $\lambda_g^0 = \lambda_{\min}(S_0(J_g))$, Weyl now yields

$$\lambda_g^0 \ge \lambda_{\min}(S_{\mathrm{rid}}(J_g)) - d_g \ge \lambda_{\min}(S_{\mathrm{rid}}(\widehat{J}_g)) - a_g - d_g. \tag{B.7}$$

Since $S_0(J_g) \succeq 0$, clipping in (10) preserves $\underline{\lambda}_g \le \lambda_g^0$. Thus monotonicity of CVaR and reciprocation give

$$\underline{I}_{\alpha,\pi} \le I^0_{\alpha,\pi} := \left[\mathrm{CVaR}_{\alpha, g\sim\pi}((\lambda_g^0 + \varepsilon)^{-1})\right]^{-1}. \tag{B.8}$$

The information-coverage conclusion follows from Appendix A.4. □

Confidence concerns held-out observations; coverage concerns the fixed weights π. A finite predeclared nuisance family can be included in the union bound over group–specification pairs; the infimum of certificates preserves coverage separately for each specification. Appendix C.4 supplies matrix allowances for frozen soft gates, including random membership normalization, pooling, and support correction, to which the same deterministic argument applies.

B.4 Implemented Audit and Numerical Comparison

Tables B.1–B.2 audit uncentered feature moments of normalized scalar task and nuisance coordinates z_i, b_i, with uniform within-group weights. Separately from the finite-sample proposition, Gaussian/Wald simulation uses 10^5 moment draws per group at *nominal* 95% confidence. For general normalized weights w_i and $u_i = (z_i^2, z_i b_i, b_i^2)^\top$, draw $m^* = (A^*, B^*, C^*)^\top \sim N(\widehat{m}, \widehat{\Sigma}_{\widehat{m}})$, where

$$\widehat{m} = \sum_i w_i u_i, \qquad \widehat{\Sigma}_{\widehat{m}} = \frac{\sum_i w_i^2 (u_i - \widehat{m})(u_i - \widehat{m})^\top}{1 - \sum_i w_i^2}. \tag{B.9}$$

Uniform weights give sample covariance divided by n. Simulation fixes realized coordinate-standardization constants and weights, omitting evaluation-fitted standardization error. Bonferroni tail $\delta_{\mathrm{W}} = 0.05/(3G)$ covers nuisance conditioning, ridge lower information, and an upper ridge gap within a run ($G = 4$ Waterbirds; 18 CivilComments).

Let J^* have blocks (A^*, B^*, C^*), and write $\widehat{F}_g^0 = S_0(\widehat{J}_g)$, $\widehat{F}_g^{\mathrm{rid}} = S_{\mathrm{rid}}(\widehat{J}_g)$. The δ_{W}-quantiles $C_g^L, L_g^{\mathrm{rid}}, L_g^0$ of $C^*, [S_{\mathrm{rid}}(J^*)]_+, [S_0(J^*)]_+$ are clipped to $[0, \widehat{C}_g]$, $[0, \widehat{F}_g^{\mathrm{rid}}]$, $[0, \widehat{F}_g^0]$, respectively. The $(1 - \delta_{\mathrm{W}})$-quantile U_g^Δ of the closed-form ridge gap is enlarged to at least its plug-in value. Lower and upper quantiles use lower and higher order statistics, respectively. The implemented lower information is

$$\underline{\lambda}_g^{\mathrm{W}} = \min\{[L_g^{\mathrm{rid}} - U_g^\Delta]_+, L_g^0, \widehat{F}_g^0\}. \tag{B.10}$$

A denominator C^* or $C^* + \lambda_{\mathrm{rid}}$ at or below 10^{-12} gives zero for its corresponding Schur value and an infinite gap; otherwise (B.4) is used even for non-PSD draws. An infinite upper gap or $C_g^L \le 10^{-12}$ forces zero and an uncertifiable status; support, empirical-PSD, and identity checks can also prevent certification. With $\lambda_{\mathrm{rid}} = 10^{-3}$, certificates with superscripts 0, rid, and W aggregate $\widehat{F}_g^0$, $\widehat{F}_g^{\mathrm{rid}}$, and $\underline{\lambda}_g^{\mathrm{W}}$, respectively, using uniform environment weights, $\alpha = 0.95$, $\varepsilon = 10^{-8}$, and fractional CVaR threshold-atom mass.

Table B.1: Same-cohort summaries of 75 default-ridge held-out audits. Conditioning/gap parentheses are within-run nominal Wald lower/upper values; entries are extrema across runs, not joint confidence statements across seeds or methods. Inflation is ridge minus exact plug-in certificate.

		Conditioning and ridge gap			Certificates	
Scope	Audits	Minimum J_{NN}	Maximum group gap	Maximum inflation	Minimum exact	Minimum Wald lower
CivilComments diagnostics	5	1.456 (1.417)	0.00116 (0.00133)	0.00018	0.4970	0.4618
Waterbirds diagnostics	5	0.211 (0.178)	0.00125 (0.00166)	0.00071	0.1639	0.1504
Frozen-feature training	30	0.227 (0.192)	0.00144 (0.00199)	0.00062	0.1972	0.1814
End-to-end training	35	0.349 (0.349)	0.00252 (0.00300)	0.00140	0.2185	0.2004
All audits	75	0.211 (0.178)	0.00252 (0.00300)	0.00140	0.1639	0.1504

Every run passed conditioning and satisfied $\underline{I}^{\mathrm{W}}_{0.95} \le \widehat{I}^{0}_{0.95} \le \widehat{I}^{\mathrm{rid}}_{0.95}$; Waterbirds diagnostic values are 0.1504, 0.1639, and 0.1646. Across the five CivilComments diagnostic seeds, the mean ridge and unregularized certificates are 0.4993 and 0.4991, respectively. The first ordering is enforced by the caps and does not establish confidence coverage. Training audits use fresh models: original training runs lacked the retained environment blocks needed for checkpoint-specific auditing.

For the separate audits in Table B.2, frozen Group DRO + FG maximizes validation certificate within two percentage points of the best validation worst-group accuracy among $\{0.1, 0.3\}$; seeds 17–21 select $(0.3, 0.1, 0.3, 0.3, 0.3)$. Repeated frozen-feature training/audit runs at ridges $10^{-4}, 10^{-3}, 10^{-2}$ give maximum group gaps 0.0001443, 0.0014374, 0.0138286 and Wald upper values 0.0001996, 0.0019880, 0.0191165.

Table B.2: Separate fresh training audits: five-seed means at the stated coefficients. Table B.1 instead includes frozen models selected over the full grid $\{0.01, 0.03, 0.1, 0.3\}$.

Training setting	Ridge plug-in	Exact plug-in	Wald lower
Frozen FG, $\lambda_{\mathrm{FG}} = 0.03$	0.4002	0.3999	0.3663
Frozen Group DRO + FG, selected from $\{0.1, 0.3\}$	0.6254	0.6252	0.5664
End-to-end Group DRO + FG, $\lambda_{\mathrm{FG}} = 10^{-4}$	0.5447	0.5442	0.4890

C Hidden-Slice Discovery

Appendix D.4 reports this protocol's outcomes. Frozen-gate evaluation recomputes membership normalization and support correction. Prompts, model revision and execution details are in the artifact.

C.1 Gate Family and Support Constraints

For the frozen standardized 576-dimensional representation $\psi(x)$ and task class c, define a logistic gate with parameters $\phi = (a_c, b_c)$:

$$h_{\phi,c}(x) = \sigma(a_c^\top \psi(x) + b_c), \qquad q_i^0 = \frac{h_{\phi,c}(x_i)}{\sum_{j \in \mathcal{D}_c} h_{\phi,c}(x_j)}, \quad \sigma(t) = (1 + e^{-t})^{-1},$$

$$m_{\mathrm{nom}} = \tau_{\mathrm{slice}} n_c, \qquad m_{\mathrm{ESS}} = \min\{n_c, \max\{1, 0.8 m_{\mathrm{nom}}\}\}, \qquad b_{\max} = \min\{1, 2/m_{\mathrm{nom}}\}.$$

Here $\mathcal{D}_c$ contains the current sample's examples with task label c, $n_c = |\mathcal{D}_c|$, and $\tau_{\mathrm{slice}} = 0.10$ is within-class mass. Discovery calibrates b_c to this mean membership; evaluation freezes the selected intercept, without re-centering or mass recalibration. The cap denominator is nominal slice size, not the ESS floor.

Support correction uses Euclidean projection of q^0 onto the capped simplex only if its maximum exceeds $b_{\max}$. Writing the result as q^{cap} and $u_i = 1/n_c$, it takes $\widetilde{q} = u + t(q^{\mathrm{cap}} - u)$ with the

largest $t \in [0,1]$ satisfying $\sum_i \widetilde{q}_i^2 \le 1/m_{\mathrm{ESS}}$. Thus $\widetilde{q} \ge 0$, $\sum_i \widetilde{q}_i = 1$, $\max_i \widetilde{q}_i \le b_{\max}$, and $n_{\mathrm{eff}} = (\sum_i \widetilde{q}_i^2)^{-1} \ge m_{\mathrm{ESS}}$. These constraints do not imply nuisance conditioning. Each sample uses its own n_c, making $\widetilde{q}$ batch-dependent. Reported mass averages raw selected-class membership; ESS uses $\widetilde{q}$.

Metrics and controls. Fisher and high-loss gates report accuracy, uncalibrated binary NLL (temperature one), and post-hoc composition averaged under $\widetilde{q}$, without information pooling. Predicted labels use a zero-logit threshold; calibrated NLL is stored separately. Random controls weight $k = \lceil \tau_{\mathrm{slice}} n_c \rceil$ sampled rows of the Fisher-selected class uniformly; the constant gate uses $1/n_c$. Unweighted and centered-unweighted controls reevaluate the predictive-Fisher and high-loss gates without rediscovery.

C.2 Discovery Objective and Settings

With matrix M_i and same-class pool $\widehat{J}_{\mathrm{pool},c} = n_c^{-1} \sum_{i \in \mathcal{D}_c} M_i$, the ideal constrained objective searches over class–gate pairs:

$$\begin{aligned} \widehat{J}^{(\eta)}_{\phi,c} &= (1-\eta) \sum_i \widetilde{q}_i(\phi) M_i + \eta \widehat{J}_{\mathrm{pool},c}, \\ (c^\star, \phi^\star) &\in \arg\max_{c,\phi} \left\{ \lambda_{\min}\left(S_{\mathrm{rid}}(\widehat{J}^{(\eta)}_{\phi,c}) \right) + \varepsilon \right\}^{-1}, \qquad 0 \le \eta < 1. \end{aligned} \tag{C.1}$$

Table C.1: Fixed hidden-discovery settings; sensitivity grids are declared before metadata reveal.

Component	Specification
Representation/nuisance	Gate: frozen ImageNet MobileNetV3-small; nuisance: CLIP ViT-B/32 image projection onto a fixed water-minus-land text direction.
Head and calibration	Binary linear head: 1,200 full-batch AdamW steps, learning rate 0.01, weight decay 10^{-4}; temperature minimizes calibration NLL over 400 log-spaced values in $[0.05, 20]$.
Gate optimization	Ten restarts; 1,000 AdamW steps, learning rate 0.03, weight decay 10^{-4}; Gaussian initial weights with standard deviation 0.02.
Matrix and stabilization	$M_i = p_i(1-p_i) v_i v_i^\top$ on two task/nuisance coordinates; $\eta = 0.10$, $\lambda_{\mathrm{rid}} = 10^{-3}$, $\varepsilon = 10^{-8}$.
Replication/sensitivity	Seeds 17–21; five validation folds; mass and shrinkage grids each $\{0.05, 0.10, 0.20\}$, headline 0.10.

The task/nuisance blocks are scalar. Optimization and checkpoint/restart/class selection use $-\log(\widehat{F}^{\mathrm{rid}} + \varepsilon) - 10(\bar{h}/\tau_{\mathrm{slice}} - 1)^2$, where $\bar{h} = n_c^{-1} \sum_i h_{\phi,c}(x_i)$. The intercept is recalibrated before discovery evaluation; only mass-feasible finite checkpoints are retained, including a constant-gate baseline. High-loss search uses the same gate, mass and support rules, replacing negative log information with weighted prediction loss. Information uses $v_i = (z_i, b_i)^\top$: z_i is the uncalibrated head logit standardized by fitting-subset moments, b_i the standardized CLIP nuisance score, and $p_i = \sigma(f_w(x_i)/T)$. Calibration changes the unnormalized predictive factor. Shrinkage pools complete matrices within the current sample's class before taking the Schur complement. Feature controls omit $p_i(1-p_i)$ but retain gate weights. The centered control forms gated and uniform same-class covariances separately, then pools them with weights $1-\eta$ and η before taking the Schur complement. For CLIP, normalize five prompt embeddings per concept, normalize their concept means, then normalize the water-minus-land difference. Its dot product with normalized image embeddings gives the raw nuisance score, without background/group annotations.

C.3 Fitting and Independent Evaluation

Table C.2 uses task labels and official splits throughout; background/group metadata remain sealed until all seeds finish held-out evaluation. Jaccard overlap uses the $k = \max\{1, \lceil \tau_{\mathrm{slice}} n_c \rceil\}$ largest raw memberships in each selected class, intersected on original row identities; different classes give zero. None of the ten Fisher/high-loss gates has a cutoff tie. Fold stability applies this rule

to the official training data's 15% calibration subset: average same-class fold-pair overlaps, then weight by selected-fold counts, excluding classes with fewer than two gates. Neither comparison uses positive-membership supports or corrected-weight ranks.

Table C.2: Fitting and evaluation sequence under the frozen protocol.

Stage	Fitted or computed	Carried forward
Training/calibration	Split official training data 85/15 by task label: fit head and coordinate normalizers on 85%; temperature on 15%. CLIP scores use official-training, then fit-subset normalization.	Fixed predictor, coordinates, nuisance direction, temperature.
Validation folds	Five shared task-stratified assignments across selectors; discover on complementary folds, evaluate on held-out folds; assess stability on the 15% training-calibration subset.	Protocol assessment; fold gates are not ensembled.
Full validation	Refit the gate under the fixed protocol; select class, gate, and restart by discovery utility.	Freeze gate weights, calibrated intercept, selected class, support settings.
Test	Apply frozen memberships; recompute admissible weights, same-class pool, and metrics before metadata reveal.	Held-out results, then post-hoc interpretation.

Support correction leaves all five primary Fisher gates' test weights unchanged; for high-loss seed 19 it raises ESS from 94.48 to 102.72. Frozen memberships need not give fixed numerical weights or inactive bootstrap correction.

C.4 Held-Out Targets and Uncertainty

Conditional on fitting data and fitted quantities, let P_c be the selected-class distribution and $n = n_c$ its evaluation sample size. For frozen gate h and matrix map M,

$$J_h = \frac{A}{p}, \quad A = \mathbb{E}_{P_c}[hM], \quad p = \mathbb{E}_{P_c}[h] > 0; \qquad \widehat{J}_h = \frac{\widehat{A}}{\widehat{p}}, \quad (\widehat{A}, \widehat{p}) = \frac{1}{n}\sum_{i=1}^{n}(h_i M_i, h_i). \tag{C.2}$$

Proposition C.1 (Frozen-gate ratio bound)**.** *For i.i.d. evaluation observations from P_c, assume $0 \le h \le 1$ and $0 \preceq M \preceq M_{\max} I_d$, with finite $M_{\max}$ specified independently of evaluation data. Given $0 < \delta < 1$, let*

$$u = \log(4d/\delta), \qquad r_A = M_{\max}\left(\sqrt{2u/n} + 2u/(3n)\right), \qquad r_p = \sqrt{\log(4/\delta)/(2n)}.$$

With conditional probability at least $1-\delta$, simultaneously $\|\widehat{A} - A\|_{\mathrm{op}} \le r_A$ and $|\widehat{p} - p| \le r_p$. Whenever $\underline{p} = \widehat{p} - r_p > 0$ on this event,

$$\|\widehat{J}_h - J_h\|_{\mathrm{op}} \le r_J := \frac{r_A + \|\widehat{J}_h\|_{\mathrm{op}} r_p}{\widehat{p} - r_p}. \tag{C.3}$$

Proof. The centered matrices $h_i M_i - A$ have norm at most $M_{\max}$ and variance bounded by $M_{\max}^2 I_d$. Matrix Bernstein [Tropp, 2012, Theorem 1.4] gives the numerator event with failure at most $\delta/2$; Hoeffding gives the denominator event with failure at most $\delta/2$. Their union bound and

$$\widehat{J}_h - J_h = \frac{(\widehat{A} - A) - \widehat{J}_h(\widehat{p} - p)}{p}, \qquad p \ge \widehat{p} - r_p > 0,$$

prove the observable bound. The shared random denominator is therefore explicitly controlled. □

Pooling and support correction. Set $J_{\mathrm{pool},c} = \mathbb{E}_{P_c}[M]$ and $J_h^{(\eta)} = (1-\eta)J_h + \eta J_{\mathrm{pool},c}$. Allocate $\delta_{\mathrm{ratio}} + \delta_{\mathrm{pool}} = \delta$, using the former in Proposition C.1. For the evaluation mean $\widehat{J}_{\mathrm{pool},c} = n^{-1}\sum_i M_i$, matrix Bernstein supplies

$$r_{\mathrm{pool}} = M_{\max}\left[\sqrt{\frac{2\log(2d/\delta_{\mathrm{pool}})}{n}} + \frac{2\log(2d/\delta_{\mathrm{pool}})}{3n}\right].$$

Writing $\widetilde{J}_h = \sum_i \widetilde{q}_i M_i$, the triangle inequality gives $\|\widetilde{J}_h - \widehat{J}_h\|_{\mathrm{op}} \le M_{\max} \sum_i |\widetilde{q}_i - q_i^0| =: r_{\mathrm{proj}}$. Hence, on the simultaneous event and when $\underline{p} > 0$,

$$\|(1-\eta)\widetilde{J}_h + \eta \widehat{J}_{\mathrm{pool},c} - J_h^{(\eta)}\|_{\mathrm{op}} \le (1-\eta)(r_J + r_{\mathrm{proj}}) + \eta r_{\mathrm{pool}}. \tag{C.4}$$

The union bound permits dependence between pool and ratio estimates; r_{proj} covers batch-dependent support correction. If the pool is fixed from fitting data, it belongs to the conditional target and needs no evaluation-error term. Appendix B.3 converts the matrix allowance into a Schur/eigenvalue lower certificate, requiring positive lower membership mass and nuisance-conditioning margins. The target retains shrinkage; evaluation-estimated centering additionally requires first-moment control.

Reported uncertainty. Each of 2,000 within-class bootstrap replicates resamples examples with membership, coordinates, logit and label together, recomputing support-corrected weights and the same-class pool. Gate coefficients/intercept, normalizers and temperature stay frozen. The 95% interval uses the 2.5th/97.5th percentiles of the pooled predictive matrix's ridge Schur value, retaining $\eta = 0.10$ and $\lambda_{\mathrm{rid}} = 10^{-3}$, with linear quantile interpolation; ε enters inverse information separately. For any reported metric T, form paired differences $\Delta_s = T_s^{\mathrm{gate}} - T_s^{\mathrm{control}}$ across five seeds, then report $\overline{\Delta} \pm t_{0.975,4} s_\Delta / \sqrt{5}$, with sample standard deviation s_Δ; single-procedure intervals use its five values. Bootstrap intervals condition on fitted quantities; seed intervals reflect shared-benchmark fitting variation. Neither validates the finite-sample radii above.

D Experimental Protocols and Additional Results

D.1 Experimental Quantities and Aggregation

Table D.1 identifies the operational targets; Appendix B.1 defines the matrix constructions. Real-data audits use nuisance ridge 10^{-3} and stabilization 10^{-8}; synthetic audits use $\varepsilon = 10^{-12}$. Certificates use $\alpha = 0.95$. Primary predefined-group coordinates use training moments; prospective normalization is specified below. Reported evaluation Loss-CVaR averages group-mean NLLs at or above their empirical 95th percentile, weighting groups equally; synthetic experiments use environment-mean NLLs. The Loss-CVaR training baseline averages the largest $\lceil 0.1n \rceil$ individual-example NLLs.

Table D.1: Diagnostic targets and replication units.

Experiment	Representation and reported quantity	Seeds
Prospective Waterbirds	Fine-tuned MobileNetV3-small; background-probe surviving fraction	22–26
Synthetic	Normalized task/nuisance coordinates; inverse-information tail and certificate	11–15
Hidden Waterbirds slices	Frozen MobileNetV3-small and fixed CLIP direction; shrunken predictive information	17–21
CivilComments groups	TF-IDF logistic classifier; feature–Schur certificate	13–17
Waterbirds groups	Frozen MobileNetV3-small and logistic fitting; feature–Schur certificate	17–21

Uncertainty. Prospective, synthetic, predefined-group, and frozen-feature training summaries report nominal 95% normal-approximation half-widths $1.96s/\sqrt{5}$, with sample standard deviation s. Hidden-slice and group-free comparisons use five-seed Student-t intervals; contrasts are paired within seed. The end-to-end frontier reports standard deviations, and the conditional Gaussian/Wald audit is separate (Appendix B.4). Hidden-gate bootstrap intervals condition on the frozen gate and recompute evaluation weights (Appendix C). Acquisition retains means without recovered per-seed intervals.

Seeds describe fitting/partition or control variation on fixed benchmark splits, not independent datasets. The frozen Waterbirds classifier and primary diagnostic are deterministic after feature extraction and logistic fitting; only its random controls and reserve anchor vary. Prospective intervals use training seeds: 23,176 composites share 5,794 birds, with two donors per bird and arm. Accuracy changes are percentage points; information retains its original units.

D.2 Prospective Nuisance Intervention

Training and frozen selection. ImageNet-initialized MobileNetV3-small [Russakovsky et al., 2015, Howard et al., 2019] is fully fine-tuned on Waterbirds [Sagawa et al., 2020] for seeds 22–26: 50 epochs, batch size 32, learning rate 10^{-4}, and bf16 on an H100. A validation-fitted temperature and class-balanced logistic background probe use the pooled representation. Five probe folds are stratified by validation group (3,000 maximum iterations); features use validation standardization. Validation nuisance scores are out of fold and use validation moments; test scores use the full-validation probe and test moments. Task logits use training moments. Test normalization never selects the slice. The rule is

$$\mathcal{G}_{\text{eligible}} = \{g : \mathrm{Acc}_v(g) \geq \mathrm{Acc}_v, \quad \mathrm{NLL}_v(g) \leq \mathrm{median}_h \, \mathrm{NLL}_v(h)\},$$
$$g^\star \in \operatorname*{arg\,min}_{g \in \mathcal{G}_{\text{eligible}}} \lambda_{\min}(\widehat{F}_g^{\text{rid}})/\widehat{J}_{TT,g}. \tag{D.1}$$

Here v denotes validation and eligibility uses uncalibrated NLL. Temperature scaling [Guo et al., 2017] minimizes full-validation binary NLL over 400 log-spaced temperatures in $[0.05, 20]$, choosing the smaller temperature at ties. The checkpoint, temperature, probe, slice and audit constants are frozen before scoring; checkpoint mismatches abort evaluation. Table D.2 shows that LL (landbird/land) and WW (waterbird/water) qualify; WW is selected in all five seeds.

Foreground-preserving construction. Waterbirds combines CUB birds [Wah et al., 2011] with Places backgrounds [Zhou et al., 2018] under soft CUB mattes. For each bird, we retain background

Table D.2: Validation selection. Means $\pm$ nominal 95% normal-approximation half-widths across seeds 22–26; accuracy is a fraction. Information uses uncentered feature moments.

(a) Performance and eligibility

Group	n	Accuracy	NLL	Eligible
Landbird/land	467	0.990 ± 0.005	0.038 ± 0.016	Yes
Landbird/water	466	0.703 ± 0.019	1.670 ± 0.156	No
Waterbird/land	133	0.390 ± 0.021	7.718 ± 0.800	No
Waterbird/water	133	0.920 ± 0.013	0.638 ± 0.038	Yes

(b) Information

Group	Raw information	Projected information	Surviving fraction
Landbird/land	0.737 ± 0.046	0.436 ± 0.045	0.590 ± 0.027
Landbird/water	0.794 ± 0.032	0.289 ± 0.014	0.365 ± 0.026
Waterbird/land	0.973 ± 0.033	0.764 ± 0.031	0.786 ± 0.036
Waterbird/water	2.063 ± 0.054	0.511 ± 0.021	0.248 ± 0.015

pixels verbatim outside the dilated foreground and fill the covered region using Telea inpainting [Telea, 2004], with a 5-pixel inpainting radius and a 5×5 elliptical dilation kernel. Its extracted bird layer is recomposed onto training-image donors of the same or opposite land/water background type. Donors have foreground fraction at most 0.08; a deterministic sequence tries four, taking the first with visible inpainted area at most 0.03, or the least-inpainted of four. The same-type arm undergoes identical recomposition.

Both arms use the same foreground layer, matte, label and source birds. Opaque foreground pixels are unchanged before JPEG encoding (quality 95, no chroma subsampling); soft-edge mixtures vary with background. Two draws per arm yield 23,176 composites. Visible inpainted area has mean 4.13%, p95 8.45%, maximum 20.37%, and arm means 4.13%/4.12%. Correctness is averaged across donors within bird, then across birds.

Outcomes and confident flips. Table D.3 reports all groups. The confident half uses the within-slice median calibrated true-label probability among originally correct birds, including ties. In WW, opposite-type/control flip rates are 56.4%/3.3% overall and 41.8%/0.7% in that half; the fixed threshold $p \geq 0.9$ selects no WW birds.

Table D.3: Held-out interventions. Accuracies and flip rates are seed means in percent; controlled drops are same-type minus opposite-type percentage points with nominal 95% normal-approximation half-widths. Panel (b) counts are mean confident-half counts.

(a) Paired intervention arms

Group	Birds	Original	Same type	Opposite type	Controlled drop
Landbird/land	2,255	98.9	99.3	72.6	26.7 ± 1.8
Landbird/water	2,255	71.3	72.3	99.2	-26.9 ± 2.0
Waterbird/land	642	42.3	40.6	92.3	-51.7 ± 2.0
Waterbird/water	642	**92.0**	**93.2**	**40.6**	$\mathbf{52.6 \pm 1.4}$

(b) Opposite-type flips among originally correct predictions

Group	All originally correct	Confident half	Confident-half count
Landbird/land	26.8	14.8	1,122
Landbird/water	0.2	0.1	805
Waterbird/land	1.2	0.3	136
Waterbird/water	56.4	41.8	299

Matched selection comparisons. Reapplying eligibility within each seed leaves LL and WW. Minimum validation accuracy and maximum uncalibrated NLL, evaluated separately on this identical

set, agree with the surviving-fraction selector in all five seeds. All-group comparisons answer a different question; landbird/water determines the model-level certificate (Table D.4).

Centering and predictive Fisher. The post hoc check fixes the checkpoints, temperatures, validation coordinates $v = (z, b)^\top$, original scales, eligible sets and intervention outcomes. No model/probe is refitted. With $\langle\cdot\rangle_g$ the divisor-n_g slice mean, compare

$$J_g = \langle vv^\top\rangle_g = \Sigma_g + \mu_g\mu_g^\top, \quad \Sigma_g = \langle (v-\mu_g)(v-\mu_g)^\top\rangle_g, \quad \mu_g = \langle v\rangle_g.$$

Centering does not rescale. The local model $\operatorname{logit} p_{\theta,\xi}(x) = f_w(x)/\widehat{T} + \theta z(x) + \xi b(x)$ at zero has score $(Y-p)v$ with conditional mean zero even when $\mu_g \neq 0$. Its predictive Fisher and the common fraction are

$$I_g = \langle p(1-p)vv^\top\rangle_g, \qquad r_\rho(A) = 1 - \frac{A_{zb}^2}{A_{zz}(A_{bb}+\rho)}, \quad \rho = 10^{-3}.$$

Perturbations follow calibration and profile only the declared background coordinate. The normalized-weight comparator is $K_g = I_g/\bar{q}_g$, $\bar{q}_g = \langle p(1-p)\rangle_g$, so $r_\rho(K_g) = r_{\bar{q}_g\rho}(I_g)$: normalization preserves the unregularized fraction but changes effective ridge. Table D.4 therefore reports it separately.

Table D.4: Selection and matrix sensitivity, evaluated within each seed. Panel (a) distinguishes original eligible and all-group sets; drops are mean percentage points. Panel (b) fixes the original eligible LL/WW set and interventions; fractions/drops are means ± nominal 95% normal-approximation half-widths.

(a) Matched selectors

Statistic	Candidates	Selected group, seeds	Drop
Minimum raw task information	Eligible	Landbird/land, 5/5	26.7
Minimum absolute projected information	Eligible	Landbird/land, 5/5	26.7
Minimum surviving fraction	Eligible	Waterbird/water, 5/5	52.6
Minimum validation accuracy or maximum validation NLL	Eligible	Waterbird/water, 5/5	52.6
Minimum normalized-weight fraction	Eligible	Waterbird/water, 5/5	52.6
Maximum NLL or minimum accuracy	All four	Waterbird/land, 5/5	−51.7
Group determining $I_{0.95}^{\mathrm{FG}}$	All four	Landbird/water, 5/5	−26.9

(b) Frozen-coordinate constructions

Construction	LL fraction	WW fraction	Selected, seeds	Drop
Uncentered J_g	0.590 ± 0.027	0.248 ± 0.015	WW, 5/5	52.6 ± 1.4
Centered Σ_g	0.779 ± 0.028	0.855 ± 0.032	LL, 5/5	26.7 ± 1.8
Predictive Fisher I_g	0.987 ± 0.014	0.273 ± 0.016	WW, 5/5	52.6 ± 1.4
Normalized weights K_g	0.987 ± 0.014	0.269 ± 0.016	WW, 5/5	52.6 ± 1.4

Centering selects LL in all seeds, whereas predictive Fisher retains WW. Slice means therefore contribute to the original ranking: for WW in seed 22, $\mu_z^2 = 2.094$ versus $\Sigma_{zz} = 0.0636$. Table D.5 preserves the per-seed reversal; full means, covariance entries, weights, matrices and hashes are retained in the artifact.

Table D.5: Per-seed fractions on the original global scales. Each cell is LL / WW; the smaller value determines selection. Centering reverses the ranking in every seed.

Seed	Uncentered $r_\rho(J_g)$	Centered $r_\rho(\Sigma_g)$	Predictive $r_\rho(I_g)$
22	0.5570 / 0.2291	0.7430 / 0.8542	0.9817 / 0.2587
23	0.6048 / 0.2494	0.8098 / 0.8481	0.9920 / 0.2729
24	0.6356 / 0.2755	0.8132 / 0.8359	0.9620 / 0.3032
25	0.5803 / 0.2392	0.7514 / 0.9156	0.9998 / 0.2571
26	0.5716 / 0.2460	0.7779 / 0.8189	0.9998 / 0.2732

Raw, projected, and tail summaries. We compare four summaries of retained validation geometry: mean raw task second moment A, arithmetic mean projected information P, reciprocal mean inverse information H, and inverse-CVaR certificate C. For the four equally weighted groups, let $F_g = \widehat{F}_g^{\mathrm{rid}}$ and $L_g = (F_g + \varepsilon)^{-1}$. Then

$$A = \tfrac{1}{4}\sum_g \widehat{J}_{TT,g}, \qquad P = \tfrac{1}{4}\sum_g F_g,$$

$$H = \left(\tfrac{1}{4}\sum_g L_g\right)^{-1}, \quad C = \left[\mathrm{CVaR}_{0.95,\pi}(L_g)\right]^{-1}, \qquad \pi_g = \tfrac{1}{4}.$$

Thus H is stabilized harmonic projected information, not mean inverse information. All four summaries use uncentered feature moments, nuisance ridge 10^{-3}, and reciprocal stabilization 10^{-8}. This post hoc analysis joins validation summaries to held-out interventions in the primary (seeds 22–26) and development (17–21) cohorts. It compares summaries of existing models, not training objectives.

Table D.6: Four validation summaries and their intervention associations. Panels (a)–(b): means $\pm$ nominal 95% normal-approximation half-widths, $1.96s/\sqrt{5}$; changes are paired within seed and drops are signed percentage points. Selected FG includes accuracy-budget fallbacks (primary seeds 22, 24; development seed 20). Panel (c): descriptive Spearman coefficients for ten models arranged in five ERM/FG seed pairs per cohort, with average ranks for tied outcomes.

(a) Model summaries and paired changes

Cohort	Model/change	A: raw	P: projected	H: harmonic	C: inverse-CVaR
Primary	ERM	1.142 ± 0.012	0.500 ± 0.010	0.442 ± 0.011	0.289 ± 0.014
	Selected FG	0.730 ± 0.086	0.484 ± 0.047	0.435 ± 0.043	0.330 ± 0.043
	FG − ERM	-0.412 ± 0.079	-0.016 ± 0.046	-0.007 ± 0.041	$+0.041 \pm 0.046$
Development	ERM	1.152 ± 0.005	0.489 ± 0.011	0.413 ± 0.014	0.255 ± 0.018
	Selected FG	0.754 ± 0.118	0.519 ± 0.065	0.433 ± 0.057	0.303 ± 0.082
	FG − ERM	-0.398 ± 0.118	$+0.030 \pm 0.063$	$+0.020 \pm 0.066$	$+0.048 \pm 0.095$

(b) ERM group-selection rules and controlled drops

Rule	Candidates	Selected, each cohort	Primary	Development
Minimum raw second moment	All four	LL 4/5; LW 1/5	$+15.8 \pm 21.3$	$+16.9 \pm 23.2$
Minimum projected / largest inverse contribution / CVaR-active group	All four	LW 5/5	-26.9 ± 2.0	-29.7 ± 2.2
Largest inverse contribution	Eligible	LL 5/5	$+26.7 \pm 1.8$	$+29.2 \pm 2.8$

(c) Model-level rank associations with held-out outcomes

	Primary		Development	
Summary	$D_{\max}$	W_{opp}	$D_{\max}$	W_{opp}
A: mean raw	-0.10	$+0.26$	$+0.26$	-0.13
P: mean projected	-0.09	$+0.21$	-0.03	-0.43
H: reciprocal mean inverse	-0.15	$+0.21$	-0.12	-0.40
C: inverse-CVaR	$+0.15$	-0.24	-0.03	-0.48

For Table D.6(c), $D_{\max} = 100 \max_g(\mathrm{Acc}_{\mathrm{same},g} - \mathrm{Acc}_{\mathrm{opposite},g})$ and $W_{\mathrm{opp}} = \min_g \mathrm{Acc}_{\mathrm{opposite},g}$, both over all four groups. Integer correct counts recover exact outcome ties: primary ERM seeds 22 and 25 share drop $348/642$, despite different floating-point subtractions. LL and LW denote landbird/land and landbird/water. Negative drops mean opposite-type recomposition improves accuracy. The largest group contributes 38.3% and 40.6% of summed inverse information on average in the primary and development cohorts, respectively. The original headline rules differed: surviving fraction in the primary cohort, absolute projected information in development; this comparison consistently uses all four groups for model summaries and correlations.

Projection and aggregation give different model summaries, with rank associations varying across cohorts. For four equally weighted groups and $\alpha = 0.95$, $C = \min_g F_g + \varepsilon$; its active group also minimizes projected information and maximizes inverse contribution. Here that group is LW, whose

opposite-type recomposition improves accuracy. The prospective WW diagnosis instead uses the surviving fraction within the eligible set. This comparison does not establish superior swap-damage prediction by inverse-CVaR; paired changes in P, H, C all have intervals containing zero. The artifact provides per-model and per-seed values, exact-count outcomes, input hashes, and replay code.

Auxiliary flip prediction. Originally correct predictions contribute two opposite-type donor rows per bird. Baseline predictors are uncalibrated NLL, calibrated true-label confidence, absolute raw logit and standardized task logit z. Augmentation adds $\beta_g b$ and $|z - \beta_g b|$, with $\beta_g = \widehat{J}_{TN,g}/(\widehat{J}_{NN,g} + 10^{-3})$ frozen from the validation block. Both models use identical five-fold, nonshuffled CUB-species `GroupKFold` splits, keeping species and donor pairs together. Logistic regression uses an intercept, ℓ_2 penalty, $C = 1$, L-BFGS, tolerance 10^{-4}, maximum 5,000 iterations, seed 0 and no class weighting or tuning. All columns, including group indicators, are centered and scaled using training-fold means and population standard deviations; standard deviations below 10^{-9} are replaced by one.

Pool out-of-fold predictions within seed before computing average precision (AUPRC), AUROC and mean log likelihood. For clipped probabilities $q \in [10^{-12}, 1 - 10^{-12}]$,

$$G_s = \frac{1}{N_s} \sum_{i=1}^{N_s} \left[\mathrm{LL}(y_{si}, q_{si}^A) - \mathrm{LL}(y_{si}, q_{si}^B) \right], \quad \mathrm{LL}(y, q) = y \log q + (1 - y) \log(1 - q). \tag{D.2}$$

G_s is augmented-minus-baseline gain in nats per donor row. Table D.7 averages paired changes equally across seeds 22–26. The coordinates improve prediction without group indicators and within WW; with indicators, AUPRC decreases. This tests their incremental value, not unique attribution to the scalar information statistic. Original inputs, exact-source refits and paired records are in the artifact.

Table D.7: Incremental flip prediction. Counts and rates are seed means; parentheses count seeds with improvement. Changes are paired within seed.

Analysis	Rows	Flip rate	Log-likelihood gain	ΔAUPRC	ΔAUROC
All groups, no indicators	9,398	0.199	+0.016	+0.035 (5/5)	+0.096 (5/5)
All groups, with indicators	9,398	0.199	+0.001	−0.007 (0/5)	+0.003 (5/5)
Selected slice	1,181	0.564	+0.013	+0.026 (5/5)	+0.015 (5/5)

Development chronology. The surviving-fraction rule followed an absolute-information pilot on seeds 17–21 and was frozen for newly trained seeds 22–26 on the same benchmark. These tables consistently use the documented primary rerun; the original-versus-rerun ledger is retained in the artifact. Appendix E.3 gives the Fisher-Glass training comparison under this same intervention.

D.3 Synthetic Classification and Tail Checks

Draw $U_e \sim \mathrm{Uniform}(0, 1)$, set $\delta_e = 0.35 U_e^{1/\gamma}$, and draw independent standard normals s, r for each observation. The task is $z = s + \delta_e r$, $Y = \mathbf{1}\{z > 0\}$, scored by nuisance-only logit $4s$. Thus

$$\Pr(\delta_e < t) = (t/0.35)^\gamma \ (0 \le t \le 0.35), \qquad J_{TT,e} = 1 + \delta_e^2, \quad F_e^0 = \delta_e^2. \tag{D.3}$$

Population feature moments are exact; loss/accuracy use 256 observations per environment. The $\gamma = 1$ experiment uses 300,000 environments and seeds 11–15. Fit the log-survival slope over the largest 5% of inverse information; empirical CVaR averages values at or above the empirical α-quantile. Figure 2 regenerates seed 11 with $L_e = (\delta_e^2 + 10^{-12})^{-1}$, rather than recovering missing original raw output; its survival curve and the five-seed summaries have different replication levels.

Table D.8 checks Theorem 1 and the scalar portfolio law in Appendix A.2. For K independent draws, information $K^{-1} \sum_j \delta_{e,j}^2$ has exponent $K/2$ at $\gamma = 1$; weights affect its constant. The mass sweep predicts $\gamma/2$. An anchor with $\delta_0 = 0.5$ and reserve mass 0.05 yields information $0.95\delta_e^2 + 0.05(0.25)$, floor 0.0125, and measured certificate 0.01260 ± 0.000001 (five seeds; Appendix E.1).

Table D.8: Synthetic checks. Panel (a) and the branch rows are five-seed means ± nominal 95% normal-approximation half-widths; zero widths reflect rounding. Mass-sweep rows are reported single-seed values.

(a) Prediction and information

Quantity	Estimate
Mean accuracy	$94.54 \pm 0.00\%$
Fifth-percentile environment accuracy	$88.67 \pm 0.00\%$
$\mathrm{CVaR}_{0.95}$ of environment-mean NLL	0.262 ± 0.000
$I^{\mathrm{FG}}_{0.95}$	$(1.22 \pm 0.53) \times 10^{-8}$
Inverse-information exponent $\widehat{\beta}$	0.498 ± 0.003

(b) Inverse-information tail exponents

Construction	Setting	Predicted β	Estimated β
Mass near the null	$\gamma = 1$	0.5	0.509
Mass near the null	$\gamma = 2$	1	0.985
Mass near the null	$\gamma = 4$	2	2.001
Independent branches	$K = 1$	0.5	0.499 ± 0.003
Independent branches	$K = 2$	1	1.000 ± 0.009
Independent branches	$K = 3$	1.5	1.508 ± 0.015

D.4 Hidden-Slice Discovery Results

Appendix C gives the blind protocol: task labels, frozen MobileNet features and a fixed CLIP direction are available; background/group metadata remain sealed through held-out evaluation. Results use the primary hidden-slice cohort, seeds 17–21, with full-validation gates frozen before test. Support correction may depend on the test sample but leaves all five primary CLIP gates' test weights unchanged. Information uses same-class pooling; accuracy, uncalibrated NLL and composition use admissible within-class weights without pooling. The artifact identifies this cohort as v2 and records original hashes; the older uncorrected cohort is excluded.

All selectors share classifiers with average accuracy $76.3 \pm 0.7\%$ and post hoc worst-group accuracy $57.3 \pm 1.3\%$. CLIP search finds 98.8% membership-weighted slice accuracy. Table D.9 includes every matched-mass comparator and paired contrast. The random-direction interval and the uncentered and centered feature-moment intervals against high-loss selection contain zero. Feature controls reevaluate gates selected using predictive Fisher without rediscovery.

Class selection and overlap. CLIP, random-direction, random-slice and constant-gate selectors choose landbirds in all five seeds; high loss chooses landbirds in seeds 17,18,21 and waterbirds in 19,20. The two different-class Jaccards are zero; same-class values are 0.00222, 0.00111, and 0. Their mean 0.000666, reported as 0.001, combines between-class and within-class differences under Appendix C.3's top-count rule.

Support and sensitivity. On same-class rows from the 15% training-calibration subset, cross-fold gate-score correlation is $0.419\,[0.353, 0.485]$ and overlap $0.258\,[0.225, 0.291]$. Test ESS is $414\,[383, 445]$ using admissible weights. Raw membership mass is $0.083\,[0.077, 0.089]$ within the selected class (target 0.10), not the full test set; landbirds contain 4,510 test examples. Validation cross-fitting, separate from final testing, gives information 0.004–0.005 over masses $\{0.05, 0.10, 0.20\}$ versus high-loss 0.012–0.027, and 0.004–0.006 over shrinkages $\{0.05, 0.10, 0.20\}$.

Composition after held-out evaluation is 97.0% landbird/land for CLIP, 94.4% for random direction, and 58.4% landbird/water for high loss. The hidden slice has lower projected predictive information than the weakest predefined group in all five seeds: mean paired difference -0.0029, with 95% Student-t interval $[-0.0034, -0.0024]$, taking the predefined-group minimum separately within each seed. Both sides use the same model, CLIP coordinate, calibrated predictive moments, pooling $\eta = 0.10$, and ridge 10^{-3}. A metadata-trained oracle probe used only after blind freeze gives information 0.003 ± 0.000, NLL 0.009 ± 0.004, and 99.1% landbird/land composition (zero width is rounding). Repair outcomes are in Appendix E.3.

Table D.9: Hidden-slice comparisons. Panel (a): means $\pm$ 95% Student-t half-widths across five paired seeds; predictive information retains calibration and shrinkage $\eta = 0.10$. Panel (b): CLIP-search minus comparator, with paired 95% Student-t intervals; negative differences indicate lower information.

(a) Held-out selectors

Selector	Predictive projected information	Uncalibrated slice NLL
Fisher search, fixed CLIP direction	0.005 ± 0.001	0.031 ± 0.017
High loss, same target within-class mass	0.019 ± 0.006	5.779 ± 0.496
Random, same target within-class mass	0.013 ± 0.001	1.025 ± 0.059
Constant class-conditional gate	0.013 ± 0.001	1.027 ± 0.049
Fisher search, random direction	0.006 ± 0.001	0.031 ± 0.015

(b) Paired information contrasts

Information	Comparison	Difference	95% interval
Predictive	High-loss slice	-0.014	$[-0.020, -0.008]$
Predictive	Random slice	-0.008	$[-0.009, -0.007]$
Predictive	Constant gate	-0.008	$[-0.009, -0.007]$
Predictive	Random direction	-0.0004	$[-0.0015, 0.0007]$
Unweighted feature	High-loss slice	-0.034	$[-0.106, 0.039]$
Centered unweighted	High-loss slice	-0.071	$[-0.166, 0.024]$

D.5 Predefined-Group Diagnostics

Fitting. CivilComments-WILDS [Borkan et al., 2019, Koh et al., 2021] predicts comment toxicity, with $y = 0$ denoting nontoxic comments. Identity-any indicates that at least one of the 24 original identity annotation scores is at least 0.5. Training-fitted unigram/bigram TF-IDF (50,000 features, minimum document frequency 5, sublinear TF) feeds class-balanced SGD logistic task/identity-any probes ($\ell_2 = 10^{-5}$, 50 passes, tolerance 10^{-3}). Identity-by-label groups require 100 examples; tail regression uses the largest 50% of group inverse information, requiring eight groups and four tail points. Seed 13 gives the fixed diagnostic; seeds 13–17 give aggregates.

Waterbirds uses frozen ImageNet MobileNetV3-small features (extraction batch 64) and balanced L-BFGS logistic task/background models (1,000 maximum iterations). Four groups are reported directly, without fitting a tail exponent; only controls/anchors vary over seeds 17–21. Primary task/probe and metadata coordinates use training moments. These original runs used an i9-11900F/32 GB workstation, about one minute for CivilComments and three minutes for first Waterbirds extraction/fitting, then cached features.

Table D.10: Original predefined-group diagnostics, distinct from fresh audits in Appendix B.4. Means $\pm$ nominal 95% normal-approximation half-widths; Waterbirds zeros reflect fixed-model fitting.

Dataset	Average accuracy (%)	Worst-group accuracy (%)	Loss-$\mathrm{CVaR}_{0.95}$	$I^{\mathrm{FG}}_{0.95}$
CivilComments	86.10 ± 0.22	43.75 ± 0.64	0.998 ± 0.012	0.500 ± 0.001
Waterbirds	77.11 ± 0.00	56.54 ± 0.00	1.741 ± 0.000	0.165 ± 0.000

For fixed CivilComments, the weakest-information group is `muslim`/$y = 0$ (accuracy 60.1%); lowest accuracy is `black`/$y = 0$ (44.0%). Waterbirds' weakest-information group is landbird/land (97.3% accuracy), whereas waterbird/land has lowest accuracy (56.5%); inverse-information p95 is 5.90.

Specification and stress controls. The prespecified semantic family is $\{b_{\mathrm{meta}}, b_{\mathrm{probe}}\}$: standardized identity-any/background annotation and its training-fitted probe, without cross-fitting. Semantic relevance follows those annotations; no held-out relevance threshold, anchor task-preservation tolerance, fold-stability admission screen, or nuisance-conditioning threshold was applied. The ridge is 10^{-3}; family minima are 0.499 and 0.165. Reserve anchors (identity-present label-balanced

CivilComments; four-group-balanced Waterbirds) are construction inputs, not admission samples. Fresh exact/ridge audits and hidden-gate stability are separate analyses.

Table D.11: Nuisance sensitivity of fixed-model diagnostics. Entries are $I^{\mathrm{FG}}_{0.95}$; worst-group accuracy remains $44.0\%/56.5\%$ for CivilComments/Waterbirds.

Specification	CivilComments	Waterbirds	Role
Metadata	0.587	0.266	Semantic-family member
Learned probe	0.499	0.165	Semantic-family minimum
Random	0.588	0.482	Negative control
Under-projected	0.588	0.482	Under-projection control
Over-projected	0.001	0.012	Task-erasure control
Hidden-loss	0.060	0.177	Loss-coordinate stress control

Random nuisance is independent standard normal; under-projection sets it to zero. Over-projection uses standardized task score plus Gaussian noise of standard deviation 0.01 and deliberately erases task information. "Hidden-loss" is observed test loss with original groups fixed. The random, over-projected, and observed-loss coordinates use evaluated-array normalization and are not independently learned nuisance scores. The metadata/random contrast is $0.587/0.588$ on CivilComments, versus probe/random $0.165/0.482$ on Waterbirds, illustrating specification dependence.

The versioned artifact (see the artifact availability statement) retains execution commands, full-precision matrices and overlaps, run-history ledgers and expanded source tables, with original cohort provenance. Those records supplement the scientific definitions and controls above.

E Information-Changing Interventions

This appendix gives the reserve proof, training and acquisition comparisons, and the influence derivative behind Section 6. Distinct cohorts and information targets remain separate. Expanded protocols and numerical breakdowns are retained in the versioned artifact's `docs/extended_appendix_details/`, with executable reproduction guides and provenance (see the artifact availability statement).

E.1 Information Reserves under Observed-Source Allocation

Fix the predictor, coordinates, and normalization. Let $J_e, J_0 \succeq 0$ have positive-definite nuisance blocks. The anchor is an additional observation source with $F_0^0 := S_0(J_0) \succeq \rho_0 I$, $\rho_0 > 0$. For the allocation in Section 6, each observation's source is observed and allocation probabilities are parameter-independent. Sampling the anchor with probability $r \in (0, 1]$ therefore gives

$$\begin{aligned} J_e^{\mathrm{res}} &= (1-r)J_e + rJ_0, \\ S_0(J_e^{\mathrm{res}}) &\succeq (1-r)S_0(J_e) + rS_0(J_0) \succeq r\rho_0 I. \end{aligned} \tag{E.1}$$

The same statement applies to affine allocation of a fixed operational matrix map; it is not an assertion about an unobserved mixture likelihood.

Proof. For $J = \left(\begin{smallmatrix} A & B \\ B^\top & C \end{smallmatrix}\right)$, $C \succ 0$, completing the square gives

$$u^\top S_0(J) u = \min_v q_J(u, v), \qquad q_J(u, v) = \begin{pmatrix} u \\ -v \end{pmatrix}^\top J \begin{pmatrix} u \\ -v \end{pmatrix}, \quad v_* = C^{-1}B^\top u. \tag{E.2}$$

Linearity in J implies

$$\begin{aligned} \min_v \{(1-r)q_{J_e}(u, v) + rq_{J_0}(u, v)\} &\geq (1-r)\min_v q_{J_e}(u, v) + r\min_v q_{J_0}(u, v) \\ &\geq r\rho_0\|u\|^2. \end{aligned} \tag{E.3}$$

The last inequality uses $S_0(J_e) \succeq 0$ and the anchor bound; holding for every u, it proves (E.1). □

For any environment distribution, $0 < \alpha < 1$ and $\varepsilon \geq 0$, CVaR monotonicity gives

$$L_e^{\mathrm{res},(\varepsilon)} \leq (r\rho_0 + \varepsilon)^{-1}, \qquad I_\alpha^{\mathrm{FG,res}}(\varepsilon) \geq r\rho_0 + \varepsilon \geq r\rho_0. \tag{E.4}$$

Here $r = m/(n+m)$ for m anchor and n base observations. An estimated anchor's positive lower bound from Appendix B.3 yields the same guarantee on its confidence event. The claim concerns the fixed predictor; retraining requires a new audit. Appendix D.3 evaluates $r = .05$, $\rho_0 = .25$, and the resulting floor .0125.

E.2 Regularized Training

For a declared group matrix $\widehat{J}_g(w)$, put $\widehat{L}_g(w) = [\lambda_{\min}(S_{\mathrm{rid}}(\widehat{J}_g(w))) + \varepsilon]^{-1}$. With fixed $\pi_g \geq 0$, $\sum_g \pi_g = 1$, training solves

$$\min_{w,\tau} \widehat{R}_{\mathrm{base}}(w) + \lambda_{\mathrm{FG}} \left[\tau + \frac{1}{1-\alpha}\sum_g \pi_g(\widehat{L}_g(w) - \tau)_+\right], \tag{E.5}$$

Here w denotes predictor parameters, $\widehat{R}_{\mathrm{base}}$ is the ERM or Group DRO task-risk objective, and λ_{FG} its information-penalty coefficient. Frozen-feature Waterbirds uses uncentered second moments of standardized task logits and fixed background-probe scores, ridge 10^{-3} and $\varepsilon = 10^{-8}$. Training $\alpha = .75$ and audit $\alpha = .95$ both give the maximum inverse information over four equally weighted groups.

Protocol and selection. ImageNet MobileNetV3-small features and background-probe scores use training-moment standardization. A linear head receives 1,200 full-batch AdamW updates (lr $= .01$, weight decay 10^{-4}, seeds 17–21), Gaussian $N(0, .01^2)$ initial weights and zero intercept. Current logits use full-training population standard deviation plus 10^{-6}; training projected information is floored at ε before reciprocal stabilization. Evaluation uses fitted training-logit moments. Group DRO minimizes the maximum group mean cross-entropy; Loss-CVaR uses the largest $\lceil .1n \rceil$ training losses, and JTT refits with weight 20 on initial ERM mistakes.

For each FG family and seed, search $\Lambda_{\mathrm{FG}} = \{.01, .03, .1, .3\}$. Among candidates whose validation worst-group accuracy is within .02 of the family's best, maximize the validation certificate, breaking ties by higher validation worst-group accuracy. This uses validation group annotations. Historical choices are .03 for ERM + FG in all seeds and .1 or .3 for Group DRO + FG. Table E.1 retains all six methods; official test data enter only final evaluation.

Table E.1: Original frozen-feature training cohort, seeds 17–21. Both panels give means ± nominal 95% normal-approximation half-widths; accuracy is percent and the uncentered feature–Schur ridge certificate uses $\alpha = .95$.

Method	Avg. acc. (%)	Worst acc. (%)	Loss-CVaR$_{.95}$
ERM	77.2 ± 0.1	55.7 ± 0.3	1.969 ± 0.007
Group DRO	80.9 ± 0.3	67.3 ± 0.3	1.143 ± 0.021
JTT	77.4 ± 0.1	54.1 ± 0.2	2.167 ± 0.004
Loss-CVaR	78.2 ± 0.1	52.1 ± 0.3	2.427 ± 0.041
ERM + Fisher-Glass	80.4 ± 0.2	56.3 ± 0.3	1.866 ± 0.025
Group DRO + FG	80.6 ± 0.9	69.4 ± 1.3	0.855 ± 0.082
Method	**Ridge certificate**	**Inverse p95**	
ERM	0.198 ± 0.001	4.86 ± 0.02	
Group DRO	0.233 ± 0.003	4.10 ± 0.06	
JTT	0.234 ± 0.001	4.13 ± 0.01	
Loss-CVaR	0.233 ± 0.002	4.17 ± 0.03	
ERM + Fisher-Glass	0.399 ± 0.006	2.46 ± 0.04	
Group DRO + FG	0.614 ± 0.059	1.63 ± 0.17	

FG doubles ERM's certificate; adding it to Group DRO raises the certificate from .233 to .614 and mean worst-group accuracy from 67.3% to 69.4%. These historical summaries differ from Appendix B's fresh ridge/exact audit cohort, which does not retroactively certify the historical checkpoints. Their original per-seed outputs were not recovered; retained reports, complete settings and independent rerun commands are in the artifact (see the artifact availability statement).

E.3 Extended Training and Selection Results

End-to-end Group DRO + Fisher-Glass frontier. MobileNetV3-small is fine-tuned for 10 epochs (batch 128, evaluation batch 512, seeds 17–21) with AdamW learning rate and weight decay 10^{-4}; the final epoch is evaluated. The reported H100 configuration uses bf16 and TF32. The nuisance is training-standardized background annotation; task logits use minibatch population standard deviation plus 10^{-6}. Groups with fewer than two observations are omitted. Ridge is 10^{-3}, $\alpha = .95$, and projected information is floored at 10^{-8} before another 10^{-8} enters the reciprocal. Table E.2 retains the complete frontier: stronger penalties can increase information while degrading prediction.

At $\lambda_{\mathrm{FG}} = 10^{-4}$ the certificate rises from .259 to .502 and inverse p95 falls from 3.68 to 2.10; at $\lambda_{\mathrm{FG}} = 10^{-2}$ worst-group accuracy is only 22.7%. Historical report summaries support this table; Appendix B's fresh audits are a separate cohort.

Prospective information training. Use Appendix D.2's shared model, 50-epoch schedule and seeds 22–26. Backbone fine-tuning uses AdamW with learning rate and weight decay 10^{-4}. FG adds coefficients $\{.003, .01, .03\}$ and a background probe trained on detached features (AdamW learning rate .001, default weight decay .01), whose scores are detached in the penalty. After one penalty-free epoch, training uses $\alpha = .75$, ridge .001, batch standard-deviation guard 10^{-6}, groups of at least six observations with nuisance standard deviation at least .05, and a .05 information floor before direct inversion. This floor differs from audit stabilization.

Table E.2: End-to-end Waterbirds Group DRO + Fisher-Glass frontier. FG entries are means $\pm$ sample standard deviations over five seeds, not confidence intervals. The Group DRO reference is the baseline from the $\lambda_{\mathrm{FG}} = 10^{-4}$ sweep; each sweep reran its baseline, so this is not a shared paired reference for every coefficient.

Objective / λ_{FG}	Avg. acc. (%)	Worst acc. (%)	Loss-CVaR$_{.95}$
Group DRO reference	85.4	53.2	0.925
5×10^{-5}	79.7 ± 6.2	56.6 ± 5.8	0.733 ± 0.061
10^{-4}	79.5 ± 10.2	57.7 ± 8.3	0.733 ± 0.110
3×10^{-4}	69.9 ± 9.7	58.8 ± 8.7	0.653 ± 0.064
10^{-3}	62.7 ± 7.9	49.9 ± 1.8	0.710 ± 0.012
3×10^{-3}	57.7 ± 4.5	48.6 ± 6.6	0.717 ± 0.036
10^{-2}	47.9 ± 16.5	22.7 ± 16.7	1.031 ± 0.416
Objective / λ_{FG}	Ridge certificate	Inverse p95	
Group DRO reference	0.259	3.68	
5×10^{-5}	0.393 ± 0.059	2.49 ± 0.34	
10^{-4}	0.502 ± 0.166	2.10 ± 0.58	
3×10^{-4}	0.614 ± 0.189	1.71 ± 0.46	
10^{-3}	0.786 ± 0.124	1.29 ± 0.20	
3×10^{-3}	0.817 ± 0.068	1.22 ± 0.10	
10^{-2}	0.809 ± 0.076	1.23 ± 0.12	

Select the final model with largest validation certificate among candidates whose overall validation accuracy is at least ERM minus .01. If none qualify, maximize the certificate over all FG candidates for that seed and record the fallback. Choices are $[.01, .03, .01, .003, .01]$, with fallbacks in seeds 22 and 24. All audit quantities are frozen before intervention scoring. In the prospective replication, FG raises the validation certificate from .289 to .330, but the controlled drop is 52.9 versus 52.6 percentage points for ERM and the originally-correct flip rate is 57.7% versus 56.4%. The accuracy constraint fails in two seeds and background sensitivity persists.

Constrained hidden-slice repair. The primary hidden-slice cohort of Appendix D.4 compares loss, Fisher and combined repair without opening group metadata. Each arm adds its repair penalties to ERM, includes the unchanged model, and must preserve overall, balanced and each-class validation accuracy within .02 of the base. Loss coefficients are $\{.1, .3, 1\}$, Fisher coefficients $\{.01, .03, .1\}$, and combined candidates use their Cartesian product. The same warm start and fresh AdamW optimizer (lr $= .01$, weight decay 10^{-4}, gradient clip 10) receive at most 1,200 steps. Calibration NLL at temperature one is checked every 25 steps, with patience 12, improvement threshold 10^{-5}, and step-zero-eligible best-checkpoint restoration. The matched ERM continuation sets both penalties to zero. Base temperature remains fixed for information training, selection and subsequent audits.

For the frozen training slice, the dimensionless Fisher penalty is

$$P_{\mathrm{FG}}(w) = \frac{\max\{F_{\mathrm{init}}, \varepsilon\}}{F(w) + \varepsilon}, \qquad \varepsilon = 10^{-8}, \tag{E.6}$$

Here $F(w)$ is the frozen slice's projected predictive information and F_{init} its value at the unchanged model, with shrinkage .1, ridge .001, and negative numerical information clipped to zero. Within the accuracy envelope, loss/Fisher repair minimizes frozen-slice loss/maximizes information. Combined selection uses the loss–information Pareto frontier, maximizing the smaller relative improvement; ties favor mean relative improvement, then the smaller coefficient sum normalized by grid maxima, $\lambda_{\mathrm{loss}} + 10\lambda_{\mathrm{Fisher}}$, then earlier index. Fresh rediscovery uses matched gate/fold seeds, so the reported population can change.

Fisher repair chooses the unchanged model in every seed. Paired worst-group changes versus matched continuation are $-.4$, .0 and $-.6$ points for loss, Fisher and combined repair, with intervals containing zero. The oracle reaches 64.8% worst-group accuracy while rediscovered information remains .005. LL denotes landbird/land.

Post hoc group-free coefficient selection. This analysis was designed after earlier group/test outcomes were inspected. Its grid $\{0, .01, .03, .1, .3\}$ uses (E.6), not the known-group objective's

Table E.3: Constrained repair on the primary hidden-slice cohort, seeds 17–21. Accuracy, NLL and information are means $\pm$ 95% Student-t half-widths; composition is a mean percentage. Panel two audits each arm's rediscovered CLIP-defined slice. The oracle uses group annotations.

Arm	Avg. acc. (%)	Worst acc. (%)	Unchanged
Matched ERM continuation	76.3 ± 0.7	57.3 ± 1.3	—
Loss repair	76.3 ± 0.6	56.9 ± 1.3	$3/5$
Fisher repair	76.3 ± 0.7	57.3 ± 1.3	$5/5$
Combined repair	76.4 ± 0.9	56.7 ± 1.6	$4/5$
Oracle Group DRO	80.0 ± 0.3	64.8 ± 0.9	—
Arm	**Slice NLL**	**Projected info.**	**LL mass (%)**
Matched ERM continuation	0.031 ± 0.017	0.005 ± 0.001	97.0
Loss repair	0.023 ± 0.017	0.005 ± 0.002	97.8
Fisher repair	0.031 ± 0.017	0.005 ± 0.001	97.0
Combined repair	0.031 ± 0.017	0.005 ± 0.001	97.2
Oracle Group DRO	0.033 ± 0.018	0.005 ± 0.002	95.9

scale. Training is partitioned $70/15/15\%$ for fitting/calibration/gate discovery; a frozen train-only slice supplies the penalty. All candidates share the warm start, training seed, optimizer, 1,200-step budget, checkpoint rule and base temperature above. Official validation enters only after heads are fixed; group metadata stays sealed through blind test evaluation.

Let $\widehat{R}_\lambda$ be mean fixed-temperature validation NLL, $s_\lambda = \mathrm{sd}_i(\ell_{\lambda i})/\sqrt{1199}$ its within-run standard error, and $\lambda_R = \arg\min_\lambda(\widehat{R}_\lambda, \lambda)$ lexicographically. Here $\ell_{\lambda i}$ is the fixed-temperature validation NLL of example i, and sd_i is the sample standard deviation with divisor 1198. Five task-stratified validation folds rediscover candidate-specific slices; $V_{k,\lambda}$ is held-out inverse predictive information and $s_{V,\lambda}$ its sample standard deviation. Selection uses

$$\begin{aligned} \Lambda_{\mathrm{risk}} &= \{\lambda : \widehat{R}_\lambda \le \widehat{R}_{\lambda_R} + s_{\lambda_R}\}, \quad U_\lambda^{\mathrm{adj}} = \overline{V}_\lambda + t_{4,.99} s_{V,\lambda}/\sqrt{5}, \\ \widehat{\lambda}_{\mathrm{hidden}} &= \min \arg \max_{\lambda \in \Lambda_{\mathrm{risk}}} (U_\lambda^{\mathrm{adj}})^{-1}, \quad t_{4,.99} = 3.74695. \end{aligned} \tag{E.7}$$

The risk allowance uses the minimizer's SE for all candidates. Bonferroni adjustment covers five coefficients, but overlapping gate-training folds make this a ranking score without independent-fold coverage. Accuracy screens and paired inverse-information tests are diagnostics, not selection constraints. The group-supervised comparator maximizes validation worst-group accuracy over the same Λ_{risk}, breaking ties toward smaller λ.

Table E.4: Post hoc coefficient selection, five paired seeds. Accuracies are test results; information is projected predictive information in test slices defined by validation-rediscovered, frozen gates. Brackets are 95% Student-t intervals. Policies selecting identical models across these seeds share rows; the text explains both coincidences. Coefficient counts are λ : number of seeds.

Policy	Average accuracy (%)	Worst accuracy (%)
Frozen ERM / matched zero	75.5 [74.3, 76.6]	55.7 [53.3, 58.2]
Fixed $\lambda = .03$	76.6 [76.1, 77.2]	51.5 [47.1, 55.8]
Risk only / group-free	77.5 [76.5, 78.6]	43.1 [33.1, 53.1]
Group-supervised comparator	77.7 [76.6, 78.8]	44.2 [34.2, 54.3]
Policy	**Coefficient counts**	**Slice information**
Frozen ERM / matched zero	— / $0:5$	0.007 [0.005, 0.009]
Fixed $\lambda = .03$	$.03:5$	0.007 [0.004, 0.010]
Risk only / group-free	$0:1,\ .3:4$	0.012 [0.006, 0.019]
Group-supervised comparator	$0:1,\ .1:1,\ .3:3$	0.011 [0.005, 0.018]

Matched zero continuation restores step zero in all five seeds, explaining equality with frozen ERM despite intervening optimization. Group-free selection equals plain NLL minimization in every seed; the information criterion does not change any choice. Versus matched continuation, paired mean changes are $+2.1$ average-accuracy points ($[+.2, +4.0]$), $+.006$ information ($[+.001, +.010]$),

and -12.6 worst-group points ($[-21.6, -3.7]$). The group-supervised comparator opens validation annotations only after blind decisions and records are fixed and is not an upper bound. The post hoc coefficient-selection results and primary repair cohort have separate artifact identifiers (see the artifact availability statement), alongside extended settings and full tables.

Operational choice of the penalty. For group-free selection, we propose prespecifying an information requirement and an ordinary-risk budget, then choosing the smallest coefficient satisfying both. If no candidate qualifies, report the requirement as unmet. With neither a specified information requirement nor a group-labeled audit set, retain $\lambda_{\mathrm{FG}} = 0$. A worst-group-accuracy objective requires group labels or a justified group proxy. This threshold-based policy requires prospective validation; the reported experiment maximizes information within an NLL eligibility set.

E.4 Tail-Transverse Influence

Fix the predictor, coordinates, calibration, slice rules and finite audit weights $\pi_g \geq 0$, $\sum_g \pi_g = 1$. An action specifies a positive-semidefinite path $J_g(t, a) = J_g + tH_{g,a} + o(t)$ as $t \downarrow 0$, with symmetric $H_{g,a}$ and an operator-norm remainder. Allocation, normalization and retained pooling belong in this path; audit weights stay fixed.

One differential for exact and ridge targets. Write $J_g = \left(\begin{smallmatrix} A_g & B_g \\ B_g^\top & C_g \end{smallmatrix}\right)$. Choose a fixed $\rho \geq 0$ with $C_g + \rho I \succ 0$: $\rho = 0$ gives exact Schur information and $\rho = \lambda_{\mathrm{rid}}$ gives the ridge target. Set $Q_g = (C_g + \rho I)^{-1}$, $S_\rho(J_g) = A_g - B_g Q_g B_g^\top$, and let λ_g be its smallest eigenvalue. Since $DQ_g[H_{NN}] = -Q_g H_{NN} Q_g$,

$$\begin{aligned} DS_\rho(J_g)[H] &= H_{TT} - H_{TN} Q_g B_g^\top - B_g Q_g H_{TN}^\top \\ &\quad + B_g Q_g H_{NN} Q_g B_g^\top, \\ p_g(u) &= \begin{pmatrix} u \\ -Q_g B_g^\top u \end{pmatrix}, \qquad u^\top DS_\rho(J_g)[H] u = p_g(u)^\top H p_g(u). \end{aligned} \tag{E.8}$$

For a simple minimum eigenvalue with unit eigenvector u_g, differentiating the eigenvalue equation therefore gives

$$h_{g,a} := D_a \lambda_g = p_g^\top H_{g,a} p_g, \qquad p_g = p_g(u_g). \tag{E.9}$$

Let s_a be the candidate's score vector and $M_{g,a}$ its matrix contribution to group g. For allocation rate $\eta_g \geq 0$, additive and normalized contributions have different derivatives:

$$\begin{aligned} H_{g,a} = \eta_g s_a s_a^\top &\implies h_{g,a} = \eta_g (p_g^\top s_a)^2, \\ J_g(t, a) = \frac{J_g + t\eta_g M_{g,a}}{1 + t\eta_g} &\implies H_{g,a} = \eta_g (M_{g,a} - J_g), \\ &\qquad h_{g,a} = \eta_g \left[p_g^\top M_{g,a} p_g - \lambda_g + \rho \| Q_g B_g^\top u_g \|^2 \right]. \end{aligned} \tag{E.10}$$

Indeed $Q_g C_g Q_g = Q_g - \rho Q_g^2$, so $p_g^\top J_g p_g = \lambda_g - \rho \| Q_g B_g^\top u_g \|^2$; the correction vanishes at $\rho = 0$. Per unit allocation, the additive rank-one contribution is the squared nuisance-adjusted residual $[u_g^\top (s_{T,a} - B_g Q_g s_{N,a})]^2$; at $\rho = 0$, this is the efficient-score residual. Replacement uses the allocated new-minus-removed contribution. For one observation appended to group g, its indicator and original sample-count scaling enter η_g; they cannot be omitted when claiming a normalized-augmentation derivative.

Tail sensitivity, including nonsmooth cases. Fix $\varepsilon > 0$ and $0 < \alpha < 1$, and put $L_g = (\lambda_g + \varepsilon)^{-1}$. Appendix A.4's finite CVaR dual uses $\mathcal{W}_{\alpha,\pi} = \{\omega \geq 0 : \sum_g \omega_g = 1,\ \omega_g \leq \pi_g/(1-\alpha)\}$. Let $\Omega_\alpha^\star(L)$ maximize $\sum_g \omega_g L_g$ over this set. If the maximizer ω_α is unique and eigenvalues are simple,

$$G_\alpha(a) := -D_a \operatorname{CVaR}_{\alpha,\pi}(L) = \sum_g \omega_{\alpha,g} \frac{h_{g,a}}{(\lambda_g + \varepsilon)^2}, \qquad D_a I_{\alpha,\pi} = I_{\alpha,\pi}^2 G_\alpha(a). \tag{E.11}$$

Thus $G_\alpha(a)$ is the first-order decrease in inverse-information CVaR along action a's specified allocation path. It follows from $D_a L_g = -h_{g,a}/(\lambda_g + \varepsilon)^2$; the dual weights include fractional threshold mass and upper-tail normalization.

For a repeated minimum eigenvalue with eigenspace $\mathcal{E}_g$, symmetric eigenvalue perturbation gives the first formula below. At a CVaR tie, the right derivative of a maximum is the maximum of its active directional derivatives; negation gives the second:

$$\begin{aligned} h_{g,a} &= \min_{\substack{u\in\mathcal{E}_g \\ \|u\|=1}} p_g(u)^\top H_{g,a} p_g(u), \\ G_\alpha^+(a) := -D_a^+ \operatorname{CVaR}_{\alpha,\pi}(L) &= \min_{\omega\in\Omega_\alpha^\star(L)} \sum_g \omega_g \frac{h_{g,a}}{(\lambda_g+\varepsilon)^2}. \end{aligned} \tag{E.12}$$

These formulas cover either or both sources of nondifferentiability; the reciprocal-certificate identity holds with right derivatives. An arbitrary fixed maximizing tail vector is only a score unless it attains this direction-dependent minimum.

For assigned label y_a, tail-transverse influence multiplies the directional information gain by current prediction loss:

$$\operatorname{TTI}(a) = G_\alpha(a)\ell_w(x_a^{\mathrm{cf}}, y_a), \qquad \ell_w(x,y) = -\log p_w(y \mid x), \tag{E.13}$$

using G_α^+ at nonsmooth points. This ranks actions at the fixed predictor; finite augmentation, refitting and changed normalization require a new audit. The implemented residual-square surrogate $\widetilde{G}_\alpha$ below omits allocation/baseline terms and is distinguished from this derivative.

E.5 Feature-Space Acquisition Experiments

Candidates, budgets and refitting. Using training-standardized frozen MobileNetV3-small representations, let $\bar{v}_{y,b}$ denote the official-training mean for task label y and background $b \in \{0,1\}$. Each official-validation candidate becomes

$$v^{\mathrm{cf}} = v + \gamma_{\mathrm{cf}}(\bar{v}_{y,1-b} - \bar{v}_{y,b}), \qquad y^{\mathrm{cf}} = y, \quad b^{\mathrm{cf}} = 1-b. \tag{E.14}$$

This preserves the assigned label, not necessarily semantic or pixel-level content. The reported unit-strength run uses label-conditional means for every candidate. A balanced logistic background probe, fitted and standardized on training data, stays fixed and scores the translated features. Test data supply final evaluation only.

Budgets $b_{\mathrm{acq}} = .01, .02, .05$ use $k = \min\{\max(1, \operatorname{round}(b_{\mathrm{acq}} n_{\mathrm{train}})), n_{\mathrm{cand}}\}$, giving $48, 96, 240$ candidates from the 1,199-example validation pool relative to 4,795 original training examples. These are formula-implied counts; historical selected-index records were not recovered. Acquired features and labels augment training, and a balanced logistic task head is fitted from scratch, without an FG penalty or special acquisition weights. Both probe and head use L-BFGS, at most 1,000 iterations, and library defaults ($C = 1$, tolerance 10^{-4}, fitted intercept). Class weights are recomputed on the augmented set. New task logits are normalized by augmented-training moments, while nuisance normalization stays fixed. Acquired validation examples are no longer an independent evaluation sample. Seeds are 17–21; the historical scikit-learn version was not pinned.

Implemented score and its scope. Let (z, q) be the task logit and nuisance score standardized on original training data. Original validation-group moments are unweighted second moments, without $p(1-p)$; put $\beta_g = J_{TN,g}/(J_{NN,g} + .001)$, $\lambda_g = J_{TT,g} - J_{TN,g}^2/(J_{NN,g} + .001)$, and $L_g = (\lambda_g + \varepsilon)^{-1}$ with $\varepsilon = 10^{-8}$. Numerical negative information is floored at zero. For the threshold-inclusive tail $T = \{g : L_g \geq \operatorname{quantile}_{.95}(L_1, \ldots, L_4)\}$, the code ranks by

$$\widetilde{G}_\alpha(a) = \frac{\sum_{g\in T}(\lambda_g+\varepsilon)^{-2}(z_a^{\mathrm{cf}} - \beta_g q_a^{\mathrm{cf}})^2}{\max\{\sum_{g\in T}(\lambda_g+\varepsilon)^{-2}, 10^{-12}\}}, \qquad \widetilde{\operatorname{TTI}}_\alpha(a) = \widetilde{G}_\alpha(a)\ell_w(v_a^{\mathrm{cf}}, y_a). \tag{E.15}$$

Every tail projector is applied to every candidate, irrespective of its assigned group; for a unique worst group the inverse-square weight cancels. The score lacks group allocation, sample-count scaling, old-information subtraction and replacement terms. Its nonnegative residual square is an unnormalized rank-one contribution at fixed coordinates, not the derivative of normalized augmentation or of the post-refit certificate. Threshold-inclusive ties also differ from the fractional/direction-dependent CVaR convention in Appendix E.4.

High loss uses ℓ_w; pure tail selects by $\widetilde{G}_\alpha$ alone. Rank product maps ascending `argsort` positions for each score evenly from 0 to 1, then multiplies their rank scores; equal values receive separate ranks. Uncertainty product multiplies $\widetilde{G}_\alpha$ by $p(1-p)$. The acquisition baseline has 78.2% average and 52.5% worst-group accuracy, certificate .209, and weakest-information group landbird/land; it is distinct from the predefined-diagnostic cohort.

Table E.5: Feature-space acquisition, means over seeds 17–21. Δa is worst-group accuracy gain in percentage points and ΔI the absolute ridge-certificate gain. Panel (a) gives the complete strength-by-budget comparison; panel (b) retains pure-tail and alternative-product controls at unit strength. Only aggregate reports survive, so historical seed-level intervals are unavailable.

(a) Strength and budget: implemented geometric score × loss versus high loss

		$\widetilde{\mathrm{TTI}}_\alpha$		High loss	
γ_{cf}	Budget	Δa	ΔI	Δa	ΔI
0.25	1%	+4.05	+0.0308	+5.76	+0.0411
	2%	+7.17	+0.0366	+7.63	+0.0477
	5%	+8.88	+0.0521	+10.12	+0.0590
0.5	1%	+6.39	+0.0205	+4.36	+0.0328
	2%	+5.61	+0.0300	+4.83	+0.0446
	5%	+6.54	+0.0417	+8.57	+0.0437
1	1%	+4.52	+0.0317	+3.27	+0.0603
	2%	+6.85	+0.0452	+5.76	+0.0581
	5%	+7.94	+0.0606	+7.01	+0.0630
1.5	1%	+7.48	+0.0662	+7.63	+0.1018
	2%	+8.72	+0.0826	+9.81	+0.1006
	5%	+9.97	+0.1159	+12.31	+0.1137

(b) Unit-strength controls

	Pure tail		Rank product		Score $\times p(1-p)$	
Budget	Δa	ΔI	Δa	ΔI	Δa	ΔI
1%	+0.62	+0.0045	+5.45	+0.0287	+1.56	+0.0012
2%	+2.80	+0.0098	+6.54	+0.0326	+1.71	+0.0073
5%	+3.58	+0.0340	+6.07	+0.0516	+3.12	+0.0230

At $\gamma_{\mathrm{cf}} = 1$, the loss product exceeds high loss in mean worst-group accuracy at all budgets, while high loss gives larger certificate gains. This advantage is strength-dependent: high loss wins accuracy at every budget for .25 and 1.5, and at the 5% budget for .5. The rank product also exceeds the chosen loss product at 1% (5.45 versus 4.52 points); the chosen product is not uniformly best. Information gains and predictive gains are distinct outcomes.

The retained acquisition reports support Table E.5; its values are not a fresh rerun. Per-seed JSON/CSV, selected indices and pinned historical versions remain unavailable, so no historical uncertainty is reconstructed. Independent rerun code, aggregate selected-candidate score/loss summaries, allocation mechanics and expanded protocol are available in the artifact (see the artifact availability statement). Five seeds on fixed splits do not constitute five independent datasets.

Declarations and Artifact Availability

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Code and supporting artifacts. The public repository and versioned release `camera-ready-v8` are available at

https://github.com/E-zClap/fisher-glass-learning/releases/tag/camera-ready-v8.

The release provides experiment code, dependencies, derived numerical inputs, per-seed records, checksums, and commands for reproducing the retained analyses and independently rerunning the experiments. Its reproduction guides specify data preparation, environments, and experiment settings. Its artifact manifest maps manuscript tables to the supporting files and identifies the corrected hidden-slice cohort. Expanded appendix sources retain displaced execution details and numerical records; the manuscript map records consolidated table labels. The prospective matrix and auxiliary flip-prediction reanalyses can be replayed from the included frozen numerical inputs. Original benchmark images and pretrained weights are not redistributed; dataset sources and access terms are documented separately. Historical synthetic and frozen-training summaries and the acquisition results retain their stated provenance: where original per-seed outputs are unavailable, the release includes the supporting aggregate reports and executable rerun workflows; fresh experiment replication does not imply exact replay of missing historical outputs.

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